

Indian Standards
by
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1. Abstract

The 2001 Gujarat Earthquake (moment magnitude 7.9) occurred near the Rann of Kachchh in northwestern India. The Gujarat earthquake demonstrates again the devastation wrought on buildings that are subjected to a major earthquake. The damage in the town of Bhuj was consistent with a felt intensity of MM X or greater. This level of energy release destroys most concrete, steel and masonry buildings within the meizo-seismal area. This paper reviews the Indian Standards for earthquake engineering, repair of buildings, plain and reinforced concrete, steel, and masonry construction. The Indian codes are compared to the United States codes of practice to determine whether any lessons can be learned from the earthquake in relation to these Indian and the United States codes of practice. The specific area of interest in the code review was the methods for estimating seismic demand and the seismic capacity. This earthquake falls in the same region as an 1819 earthquake. There was a short repeat time between events. Should be considered in code development?

2. Introduction

The purposes of this paper were threefold, to provide an overview of current codes, to illustrate the seismic hazard maps for India, and to show the methods for estimating seismic demand and seismic capacity within the codes.

The late Prof. Jai Krishna of the University of Rookee headed the development of the earthquake engineering codes of practice issued by the statutory institution the Bureau of Indian Standards. The standards that are applicable in the earthquake-engineering field in India are IS 13935: 1993 Repair and Seismic Strengthening of Buildings – Guidelines, IS 1893: 1984 Criteria for Earthquake Resistant Design of Structures, IS 456: 2000 Plain and Reinforced Concrete Code of Practice (Fourth Revision, IS 13920: 1993 Ductile Detailing of Reinforced Concrete Structures Subjected to Seismic Forces - Code of Practice, IS 800: 1984 Code of Practice for General Construction in Steel (Second Revision) and IS : Masonry Structures.

3. Seismicity and Damage

The interesting feature of this earthquake was the repeat damage to structures repaired after the 1819 earthquake (Figure 1). Investigators have made similar observations after recent eastern Mediterranean earthquakes. This point on repeated damage was raised after the 1989 Newcastle



Figure 1 Collapse of Historic Wall in Ahmedabad

¹ My section of the report – full report at Mid America Earthquake Center

earthquake and recently for the Charleston, South Carolina area with the public buildings repaired after the 1886 Charleston earthquake. This problem of the repeated damage raises the question as to the design life for some structures and the accumulation of damage in structures that are subjected to multiple earthquakes such as occurs in California (Nichols, 1999; USGS, 1977; Algermissen, 1972).

The town of Bhuj in the State of Gujarat suffered almost complete destruction in the 2001 Gujarat earthquake (Figure 2.) There was little doubt that this damage level represents a felt intensity MMI X to XII zone. The high death toll suggests that it was MMI XI+ (Shiono, 1995).



Figure 2. Damage at Bhuj, Gujarat State

IS 1893: 1984 Criteria for Earthquake Resistant Design of Structures was the standard used for the earthquake design of structures in India. The Sectional Committee who prepared the standard made a number of observations that are relevant to this work. These observations were:

'It is not intended with this standard to lay down regulations so that no structure shall suffer any damage during earthquakes (sic) of all magnitudes. It has been endeavored to ensure that as far as possible structures are able to respond without structural damage to shocks of moderate intensity and without collapse total collapse to shocks of heavy intensity.' This condition is in line with the methods being developed through the Federal Emergency Management Agency for the seismic designs in the United States (FEMA 273, 1997).

'Though the basis for the design of different types of structures is covered in this standards, it is not implied that detailed dynamic design analysis shall be made in every case. There might be cases of less importance and relatively smalls structures for which no analysis need be made, provided certain simple precautions are taken in the design... Similarly in highly seismic areas, construction of a type which entails heavy debris and consequent loss of live and property such as masonry, particularly mud masonry and rubble masonry should be avoided in preference to construction of a type which is known to withstand seismic effects better, such as construction in lightweight materials and well braced timber-framed structures.'

This frank statement of intent of the code committee recognizes the problems of poor quality masonry construction in highly seismic zones. The continuing difficulty that the code committee grapples with is the political reality of the existing construction and the economics of future construction. The Code Committee had no intention of permitting the inadequate design of structures in the higher seismic zones, although this was not expressly stated.

The Committee defined the seismic zones using five categories numbered I to V. Figure 3 shows the seismic zone map for the State of Gujarat reproduced from the code's Figure 1 (IS 1893: 1984, pg 7-8 and Appendix E). The key cities in the area are Bhuj that was clearly Zone V, and Rajkot and Ahmedabad that were clearly in Zone III.

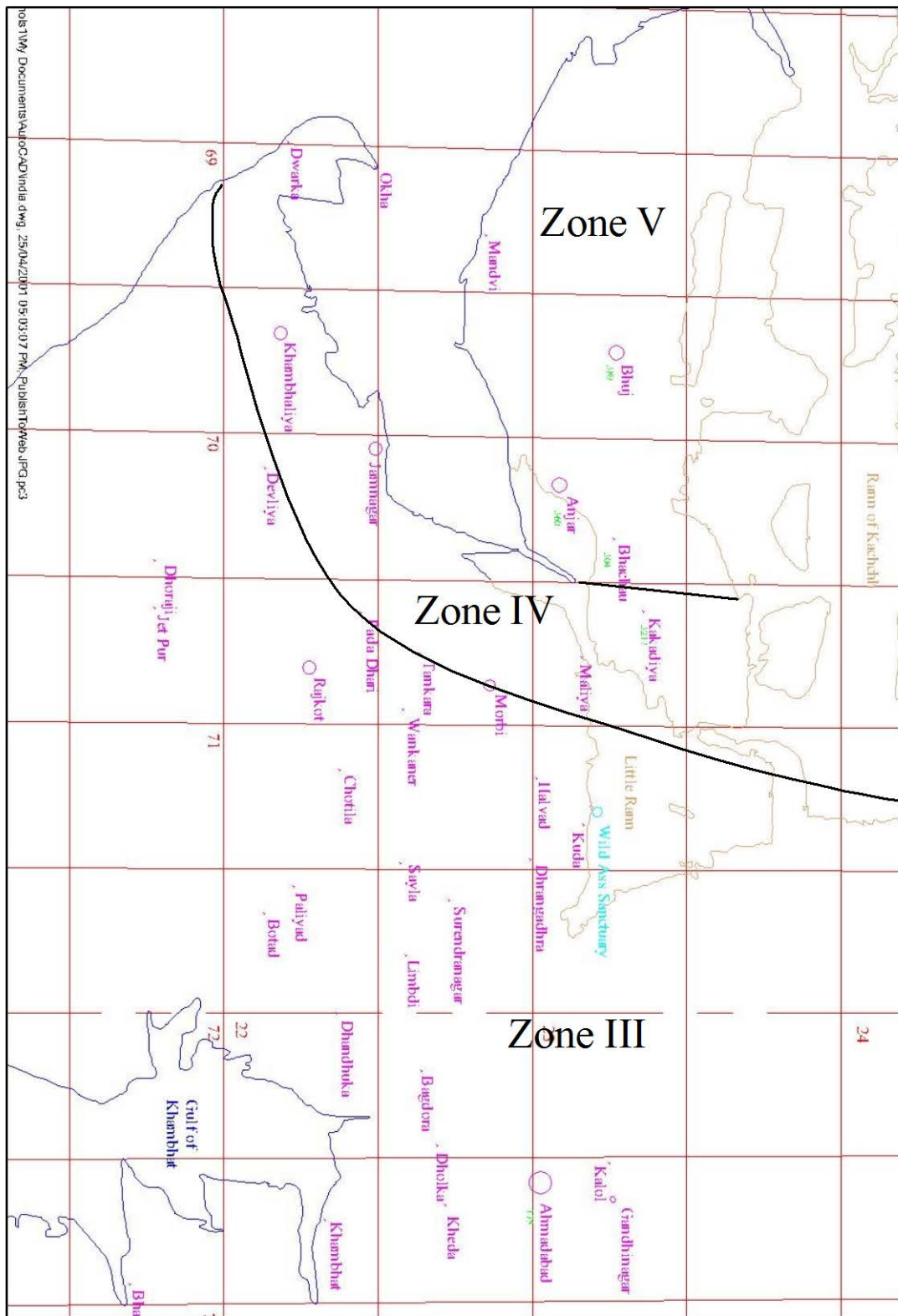


Figure 3. Gujarat State Seismic Zonation

The committee provided the following correspondences between the Modified Mercalli Intensity and the Indian Zonation (Table 1).

Table 1. Definition of the Indian Zonation Standard

Zone	Mod. Mercalli Intensity	Basic Horizontal Seismic Coefficients α_0	Town (Distance to epicenter)	Seismic Zone factor F_0 for average acceleration spectra	Abrams' Estimate of the MMI ²
I	V or less	0.01		0.05	
II	VI	0.02		0.10	
III	VII	0.04	Ahmedabad (255)	0.20	VI
IV	VIII	0.05	Morbi ³ (100)	0.25	VII
V	IX to XII	0.08	Bhuj (40) Dudhai (14)	0.4	IX at 40 kms and XII at 12 kms.

The attenuation of the felt intensity for the Gujarat earthquake was not as rapid as observed in Tangshan for equivalent distances. The difference may be due to a difference in construction standards, ground conditions and at distance in Tangshan the ground has thinner sediments over coal bearing rock. The MMI data collated by Abrams has been plotted against the Tangshan data collated by Shiono (Figure 4). A polynomial has been fitted through the Gujarat data. The point at 130 kilometres will require further investigation.

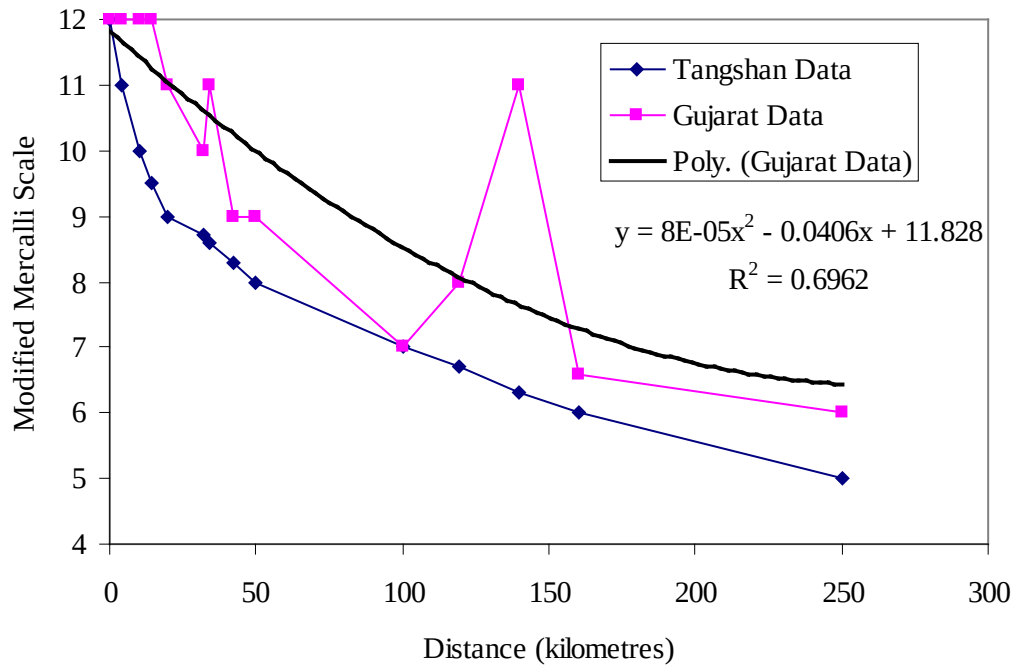


Figure 4 Gujarat and Tangshan attenuation of Felt Intensity

² Chapter 2?

³ Interpolation of the Indian Standard figure would appear to place Morbi in Zone IV.

Rossetto (Chapter 9) noted *‘that together with the increased damage potential of the ground motion due to possible site amplification and earthquake duration effects, bring the structures to exhibit a higher vulnerability than would be expected in a population of typical European buildings.’* The difficulty in reaching conclusions based on this limited data set was in determining what were the problems.

1. Are the buildings more vulnerable than equivalent European and American buildings?
2. Or has the earthquake energy attenuated at a lower rate than for the equivalent European setting?
3. Did a lower attenuation rate couple with the duration of the earthquake to cause the increased damage?

The rapid attenuation in the 1915 Avezzano earthquake in Italy in the east west direction underscores the difference in the interplate and intraplate areas (Nichols and Beavers, 2000). The attenuation rate in the New Madrid Seismic Zone and the Gujarat State will need to be the subject of further study.

The clear conclusions that can be reached from the examination of the Indian Earthquake Standard are that the Code Committee has adequately addressed the question of estimating the likely intensity of shaking, at least from the perspective of the Modified Mercalli Scale. The code has addressed the issues of minor buildings in a sensible manner, although one could suggest that the note was subject to interpretation. The code committee has recommended against the common practice of poor quality masonry construction, which conforms with the very specific and detailed conclusions drawn by Abrams (Chapter 6).

The issue of the intraplate attenuation rate that has arisen in the last few years with the research on the New Madrid Seismic Zone and from the Tangshan data strongly suggests that this point will continue to require significant and ongoing research in India, China, and the United States. The economic impact from failure to address this issue in the central and eastern United States will be significant. The Standards committee would have been unaware of the magnitude of this problem at the time of preparation of the standard.

4. Seismic Code (IS 1893: 1984 Criteria for Earthquake Resistant Design of Structures)

The IS 1893: 1984 Criteria for Earthquake Resistant Design of Structures has been compared to the requirements of FEMA 273 and 356 for the design of structures subjected to earthquake loading.

Three source documents for the development of the Indian and American standards are used in this comparison (Newmark and Hall, 1968, 1978; Veletsos and Newmark, 1964). The original documents are used to confirm the applicability of the selected methods and whether change or additional research is warranted. The work by Veletsos and Newmark (1964) includes pulse waves, which are of interest in near field analysis for great earthquakes (Richter, 1958) and Fast Fourier transform methods (Brigham, 1988).

Veletsos and Newmark⁴ (1964) noted the limited seismic data of the time. The FFT procedure was not available to these researchers, although they use the Fourier mathematics in their analysis.

IS 1893: 1984 provides the design criteria for multi-storied buildings that was based on the estimation of the equivalent static base shear:

$$V_B = KC\alpha_h W \quad (1)$$

where W represents the dead load and a fraction of the live load depending on the Load Class being either 25 percent or 50 percent; α_h was the design seismic co-efficient; C represents the flexibility of the structure related to the number of storeys and the fundamental period T ; K represents the performance factor depending on the structural framing system and the ductility level for construction; V_B represents the base shear load on the structure.

The fundamental period was estimated for moment resisting frames without bracing or shear walls using:

$$T = 0.1n \quad (2)$$

where n defines the number of storeys. The equation for all other structures was:

$$T = 0.09H / \sqrt{d} \quad (3)$$

where H represents the height of the building in metres; d represents the maximum base dimension in metres in the direction parallel to the applied seismic load.⁵

The equation provides an estimate that was equivalent to a Finite Element analysis of a degrading structure on a soil. The equation for the frequency f was:

$$f = \frac{10}{n} \quad \text{or} \quad f = 11\sqrt{d}/H \quad (4)$$

Figure 5 presents factor C determined from the IS graph (IS 1893: 1984, pg 22).

⁴ These documents use the frequency as the standard unit for expressing dynamic structural response. This method will be used in this document to be consistent with the work of Newmark and to avoid the inversion problem associated with use of period. The seismic community records earthquakes in the time domain and reports results as a function of time and frequency. Inversion to period introduces a distortion to the graphing of the results. Whilst the conversion is trivial its use should be discouraged in code documents and in the education of young engineers, who will increasingly work directly with seismic data and Fast Fourier transforms.

⁵ Equation (3) estimates a period for a 20-metre masonry tower with a base dimension of 4.5 metres at 0.84 seconds or a frequency of 1.2 Hertz, and for a 15-metre structure at 0.46 seconds or 2.1 seconds.

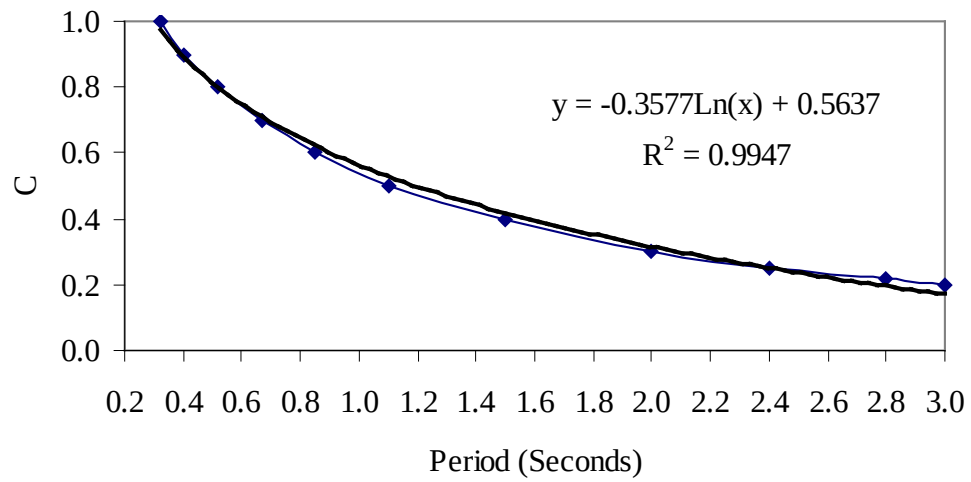


Figure 5. The Factor C against Period T

The data has been expressed in terms of frequency, as that was relevant for masonry structures (Figure 6).

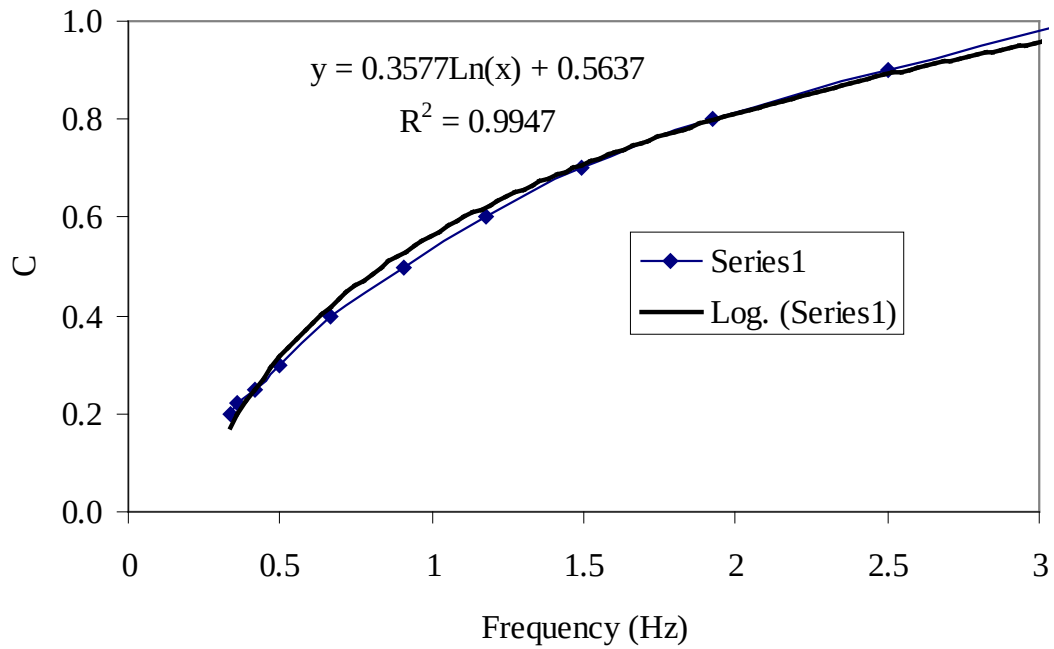


Figure 6. The Factor C against Period F

This result shows that the code provides a higher base shear for shorter structures such as low masonry buildings.⁶ This results implies that:

$$C = (0.3577 \ln(f)) + 0.5637 \quad (5)$$

The Indian equations for determining the period can be traced to ATC 3-06 (1978) except that the moment resisting frame (steel and concrete) in ATC 3-06 had the equation for period of:

$$T = C_T h_n^{0.75} \quad (6)$$

where the values for C_T for ATC 3-06 and the values from FEMA 302 (1997) are presented in Table 2.

Table 2. Coefficients for the determination of period of a Structure

	C_T			Period	Frequency
Frame Type	ATC 3-06	FEMA 302	Metric ⁷ co-efficient	20 metre height	20 metre height
Steel – moment resisting	0.035	0.035	0.0853	0.80	1.23
RC – moment resisting		0.030	0.0731	0.69	1.44
Reinforced Concrete	0.025		0.0609	0.57	1.73
Eccentrically braced steel frame		0.030	0.0731	0.46	2.16
All other framing systems		0.020	0.0488	1.38	0.72
Wood frame (2 to 3 storeys)		0.060	0.1462	0.80	1.23

FEMA 302 (1997) allows the use of equation (2) provided that the buildings was less than or equal to twelve stories and has minimum storey height of 3 metres. A finite element analysis of the earlier detailed masonry tower provided a period of 0.46 seconds and a frequency of 2.2 Hz. This result represents a higher frequency or lower period that the results calculated using equation (3). Chopra and Goel (2000) comment in detail on the derivation of these types of equations and the intentions of the code committees.

The selection of the factor C depends on the calculated frequency. There is a conceptual difficulty in selecting a bounding function such as equation (2) or (6). The form of two equations depends on the researchers original data set, the selection of the type of the fitted equation and the fitting method⁸ used in the analysis. The equations give reasonably comparable answers. Chopra and Goel (2000) provide a range of solutions to this problem of selecting a natural frequency equation. The issue of building response frequency will need to be reviewed in both the IS and FEMA 302 based on the recent work of Chopra and Goel.

⁶ The example masonry tower would have a C of 0.6.

⁷ These metric co-efficients are used in the nomenclature for the subsequent figures.

⁸ The three fitting methods could be the least squares, robust or bounding functions. The least squares provides a mean answer that can be distorted by extreme outliers, the robust has the problem of equal weight for points and the bounding will introduce systematic errors as identified by Chopra and Goel.

Two methods are provided in the IS 1893: 1984 for calculating the design seismic co-efficient. The first method was the seismic co-efficient method (equation 7a) and the second is the response spectrum method (equation 7b).

The forms of the equations for the two methods are:

$$\alpha_h = \beta I \alpha_0 \quad (7a)$$

$$\alpha_h = \beta I F_0 \frac{S_a}{g} \quad (7b)$$

where β defines different soil-foundation systems (Table 3) and $\frac{S_a}{g}$ was defined from an average acceleration spectra (Figure 7), and I defines the importance factor (Table 4).

Table 3. Values for β for Different Soil-Foundation Systems⁹

Sl No.	Type of soil mainly forming the foundation	Piles passing through any soil but resting on Soil Type 1	Piles not covered under Column 3	Raft Foundations	Combined or isolated RCC Footings with tie beams	Isolated RCC Footings without tie beams or unreinforced strip foundations	Well foundations
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i	Type I Rock or hard soils	1.0	-	1.0	1.0	1.0	1.0
ii	Type II Medium Soils	1.0	1.0	1.0	1.0	1.2	1.2
iii	Type III Soft Soils	1.0	1.2	1.0	1.2	1.5	1.5

Table 4. Values for I for Structures

SL No.	Structure	Value of Importance Factor
(1)	(2)	(3)
i	Dams (all types)	3.0
ii	Containers of inflammable or poisonous gases or liquids	2.0
iii	Important service and community structures, such as hospital, water towers and tanks, schools, important bridges; important power houses; monumental structures; emergency buildings like telephone exchanges and fire brigade; large assembly halls and subway stations,	1.5
iiii	All others	1.0

⁹ The value of β for dams shall be 1.0

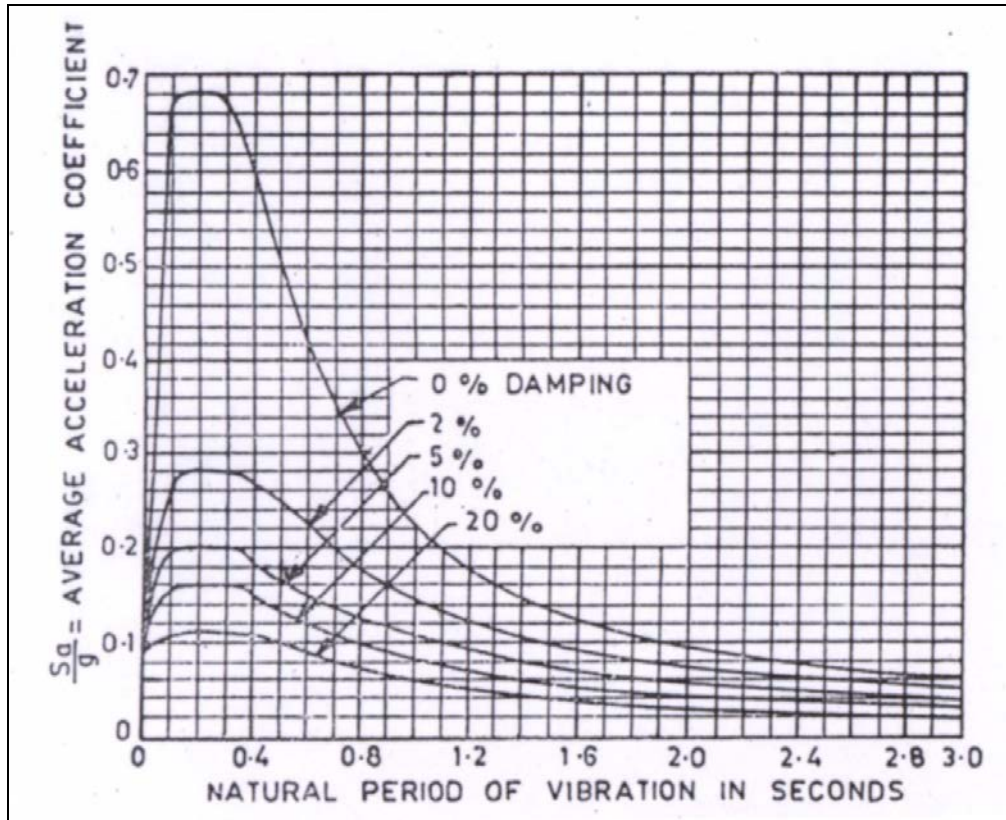


Figure 7. Average Acceleration Spectra (after IS 1893: 1984, Fig. 2, pg 18)

K represents the performance factor depending on the structural framing system and the ductility level for construction (Table 5).

Table 5. Values for K for Structures

SL No.	Structural Framing System	Value of Factor	Remarks
(1)	(2)	(3)	(4)
i (a)	Moment resistant frame with appropriate ductility details as given in IS: 4326-1976	1.0	--
i (b)	Frame as above with RC Shear walls or steel bracing members designed for ductility	1.0	These factors will apply only if the steel bracing members and the infill panels are taken into consideration in stiffness as well as lateral strength calculations provided that the frame acting alone will be able to resist at least 25 percent of the design seismic forces.
ii (a)	Frame as in (i) (a) with either steel bracing members or plain or nominally reinforced concrete infill panels	1.3	
ii (b)	Frame as in (i) (a) in combination with masonry infills	1.6	
iii	Reinforced concrete framed buildings (Not covered above)	1.6	

The distribution of forces in the code uses:

$$Q_i = V_B \frac{W_i h_i^l}{\sum_{j=1}^{j=n} W_j h_j^l} \quad (8)$$

where Q_i represents the lateral forces at the roof of floor i , W_i represents the dead and live load (portion) of the roof and any floor i , *the weight of the walls and columns in any storey is assumed to be shared half and half between the roof or floor at top and the floor or ground at the bottom, and all weights are assumed to be lumped at the level of the roof or any floor i .* h_i represents the height measured from the base of the building to the roof or any floor i and n represents the number of storeys including the basement floors, where the basement walls are not connected with ground floor deck or the basement walls are not fitted between building columns, but excluding the basement floors where they are so connected. l represents an exponent that for this IS code was two for all periods.

The original development of this force distribution in ATC 3-06 (1978) has a variable for l (Table 6).

Table 6. Values for l for Structures

Period (Seconds)	Frequency (Hz)	Value of l
0.25	4	1
0.5	2	1
1.0	1	1.25
1.5	0.67	1.50
2.0	0.5	1.75
2.5	0.4	2
5	0.2	2
10	0.1	2

The effective quasi-static limit for masonry has been estimated from experiment and a theoretical analysis of the errors in Newton's equation of motion at 0.4 Hz (Nichols, 2001).

The Average Acceleration Spectra provided in the Indian Standard (Figure 7) has been converted to the frequency spectra for the two percent damping (Figure 8). The relative paucity of points in the one to ten hertz region was considered to reflect the significant problem of translating the seismic data from the frequency domain into the inverted period form. The use of the period represents a carry-over from the twenties and electric circuit theory (Hall, 2001).¹⁰

One earthquake that has been used in developing the seismic standards for the CEUS was the Nahanni Earthquake¹¹ (Figure 9). This M6.8 event occurred in the Northwest Territories. The Nahanni earthquake has a series of spikes in the acceleration record. Veletsos and Newmark (1964) in the analysis of shock loading generated by bomb blast have considered the analysis of such pulse loading. This method is considered applicable to these large short duration pulse loads.

¹⁰ Hall, W, (2001) personal conversation dated 26 April 2001 on the development of the period formula.

¹¹ Station 1, Iverson, Canada: 23 Dec, 1985

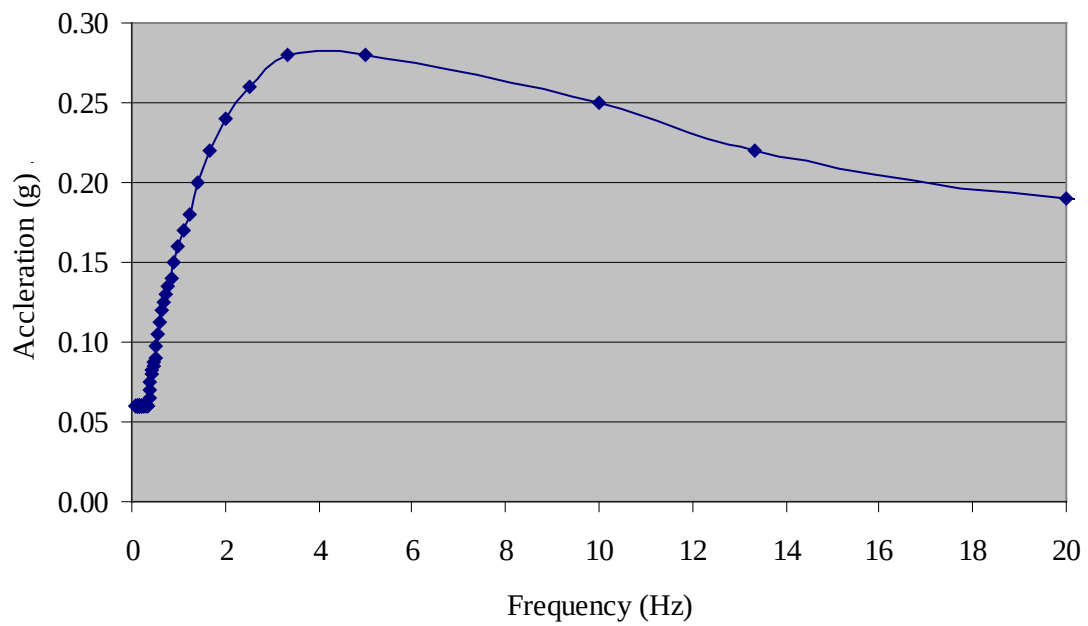


Figure 8. Average Acceleration Spectra in frequency domain

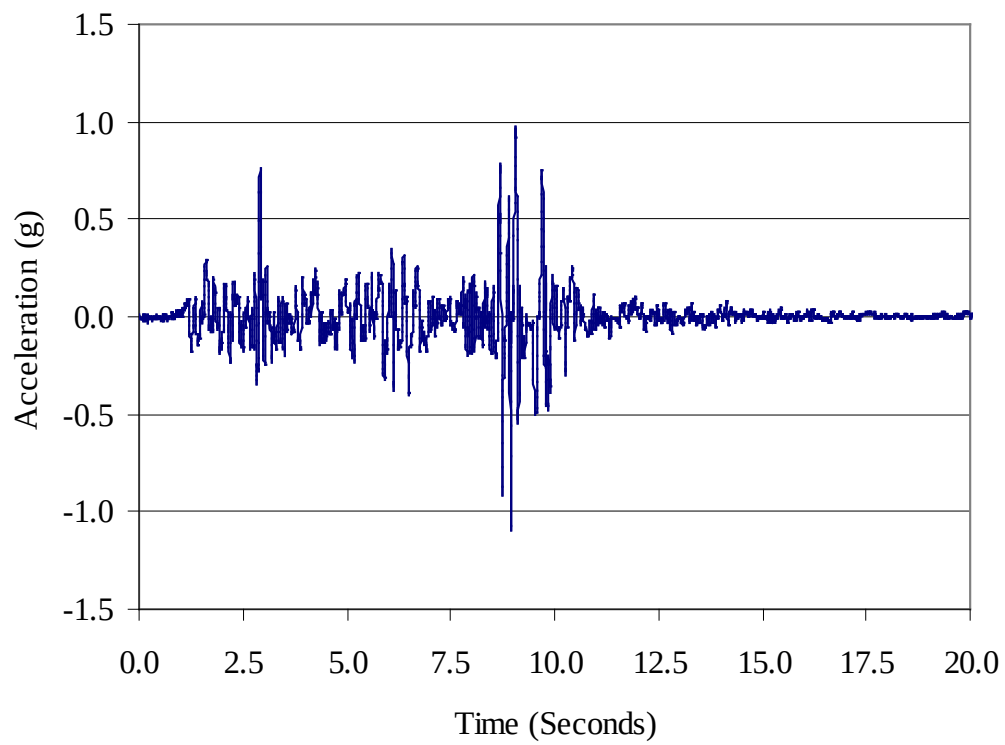


Figure 9. M6.8 Nahanni earthquake longitudinal time trace

The work of Atkinson and Boore (1995) for Eastern North America (ENA) provides one interesting seismic data set. The two specific earthquake events that were used in this work, were at Saguenary, Quebec (1988) and Nahanni, Northwest Territories, both located in Canada. The new ground motion relations provided by Atkinson and Boore cover the peak ground motions and response spectra for Eastern North America events in the range of M 4 to 5. Atkinson and Boore noted that this analysis was consistent with the data from the Nahanni and Saguenary earthquakes. Whilst the trace (Figure 9) was instructive, it does little to provide guidance as to the frequency relationship for the earthquake trace.

This data on the frequency component is instructive when it is matched with building data and the modal frequencies. The pulse analysis technique of Veletsos and Newmark (1964) can be applied to the low frequency peaks in the FFT. The same trace (Figure 9) was transformed using the procedure from Numerical Recipes in Fortran¹² in a program named CONVERT. The discrete Fourier transform programs were checked using the exponential function routine and the square wave routine from Brigham (1988, Chapter 9). The resultant frequency domain trace for the time-domain acceleration trace has been completed with frequency as the dependent variable (Figure 10) and using the period as the dependent variable (Figure 11). Note the point density differences in Figure 11, the fact that one third of the graph represents the quasi-static region; one third represents the region from 0.4 to 1.0 Hz and one third the region from 1.0 Hertz to the recorded limit. The FFT limit from typical time traces is 25, 40 or 100 Hertz.

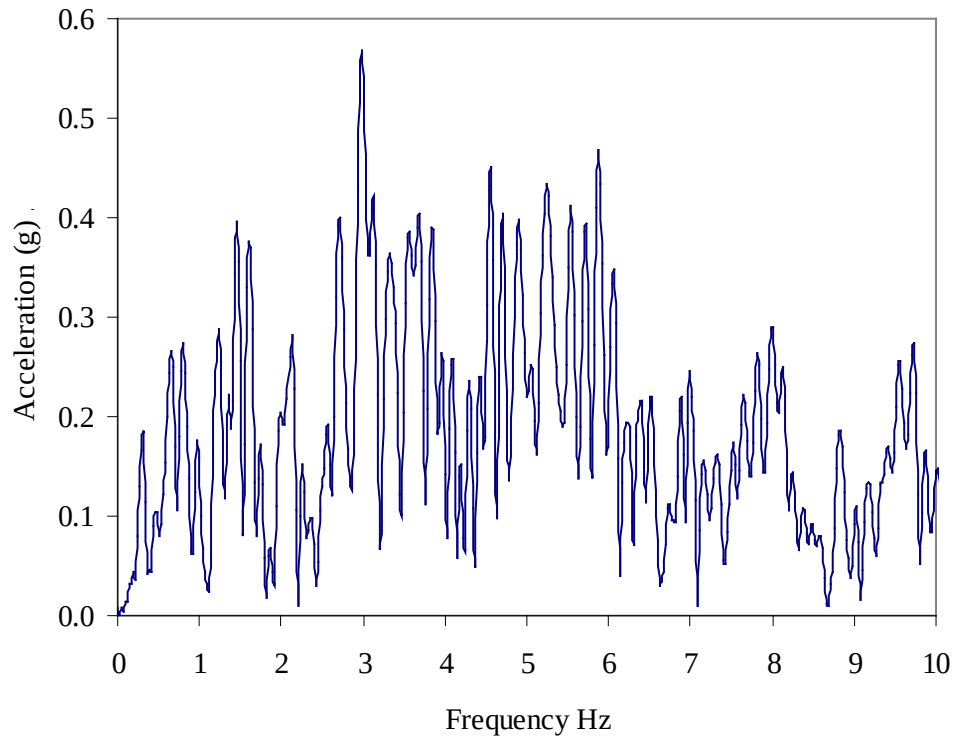


Figure 10. FFT of the Nahanni earthquake.

¹² Press et al.,(1994), *Numerical Recipes in Fortran* (Cambridge: Cambridge UP.), DFOUR Routine

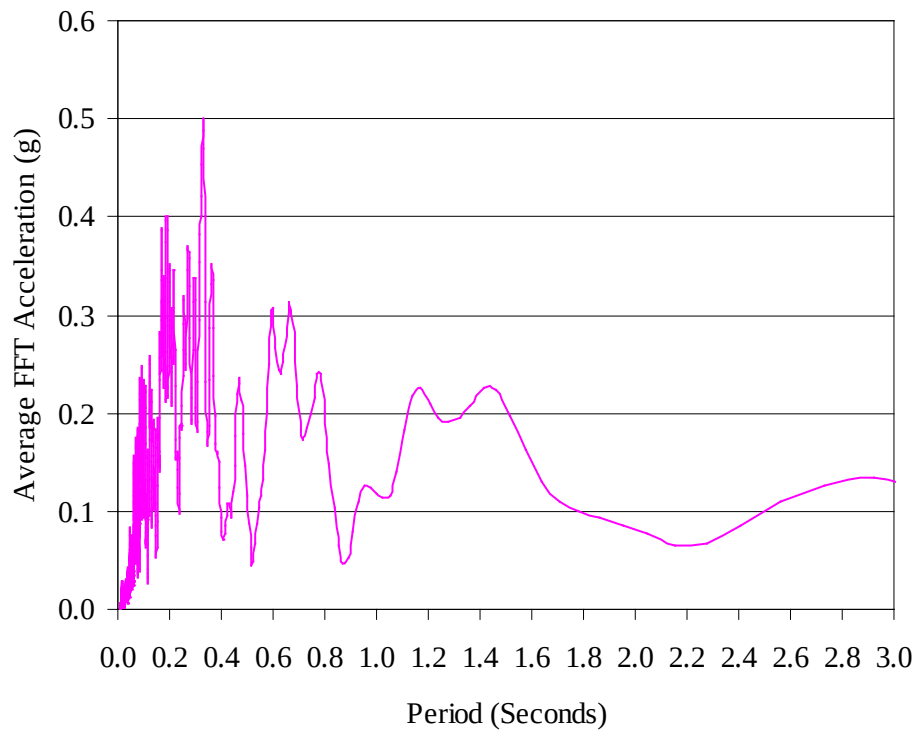


Figure 11. FFT of the Nahanni earthquake (period)

At Memphis, one has access to the work of Horton, Barstow, and Jacobs¹³ who have generated synthetic earthquakes of varying M_r combinations and stress drops for the LAMB project (Abrams and Shinozuka).¹⁴ Jacobs¹⁵ noted that in the production of these records that ‘*natural records are always ‘holey’ or peaked and troughed, but to various degrees. This spectral roughness is pronounced when you make Fourier spectra or undamped response spectra, but diminishes of course with increasing damping of the SDoF resonator for response spectra. The latter is what happens really in damped structures. Occurrence of spectral roughness is a strong function of the site conditions: i.e. of the impedance (shear-wave velocity times density) profile with depth for the site. The Nahanni and El Centro records have less contrast-rich soil/rock profiles, while the sites for Memphis we modelled have very soft soils overlying denser sands, and eventually hard rock, therefore being very ringy and full of holes and peaks. We start out in the synthetics with a smooth Brune spectrum, which becomes a bit more ringy after the crustal wave-propagation scattering function is applied (and the Earth does that also, that’s why we mimic it); but once we add in the soil response we really get site-specific ‘holiness’. That is the reason why you sometimes get surprising damage for certain structures at certain sites: double resonance between site peak and fundamental period of the structure. The most famous case for this is of*

¹³ Horton, S.P., Barstow, N., and Jacobs, K., (1997), Simulation of earthquake ground motion in Memphis, Tennessee, Proceedings of the Eleventh World Conference on Earthquake Engineering, June 23-28, 1997 Acapulco, Mexico, Elsevier Science, Paper 1302, passim.

¹⁴ Abrams, D.P., and Shinozuka, M., (1997), Final Report Loss Assessment of Memphis Buildings, Technical Report NCEER-97-00, (Urbana, Illinois: UIUC), passim.

¹⁵ Jacobs, K., Lamont Doherty Observatory, Columbia University, in personal communication in November 1997.

course the Mexico City effect in 1985 (?): 1- 2 second 10-20 story high rise buildings located on the clayey Old Lake bed (site resonance's peaked near 1 - 2 seconds) collapsed, while people in adjacent old colonial masonry buildings 2 or 3 stories high did not even realize there was an earthquake (totally detuned). If you are interested in a generic study, then you must use several of the records from different site conditions, and average the effects, so that you never have structural performance for a given building height or fundamental period influenced by the peaks or troughs of just one site condition. I do not think that the use of artificially smoothed spectra is a valid substitute. Rather the smoothing of performances over real site conditions seems to me a more valid approach, at least if you are not only interested in the mean response, but also want to get a measure for the variability of response, which highly depends on whether double resonance's of sites and structures do or do not occur under region-specific site conditions and their respective variability. The Nahanni earthquake tolerably matches the undamped average acceleration spectra from the Indian Standard 1893: 1984. The Nahanni earthquake has a spike of 1.5 g's and a short time at about 12-15 seconds. The Gujarat earthquake was a larger, longer event. The Indian Standard 1893: 1984 was based on the work from ATC 3-06 (1978) that was largely developed at the University of Illinois by Newmark, Hall and Souzen. The current FEMA documents related to the design of new structures are derivatives of the ATC manual. The two standards IS and FEMA have diverged in the base equations that have been selected from the ATC work. It was instructive to compare the calculated results from the two derived documents for a typical range of building sizes. The building has been assumed to have three depths ten, twenty and thirty metres and heights ranging from three metres in increments of three metres. The analysis was completed for Zone V, or MMI XI-XII. The building period was determined against the height for the varying type of structure (Figure 12).¹⁶

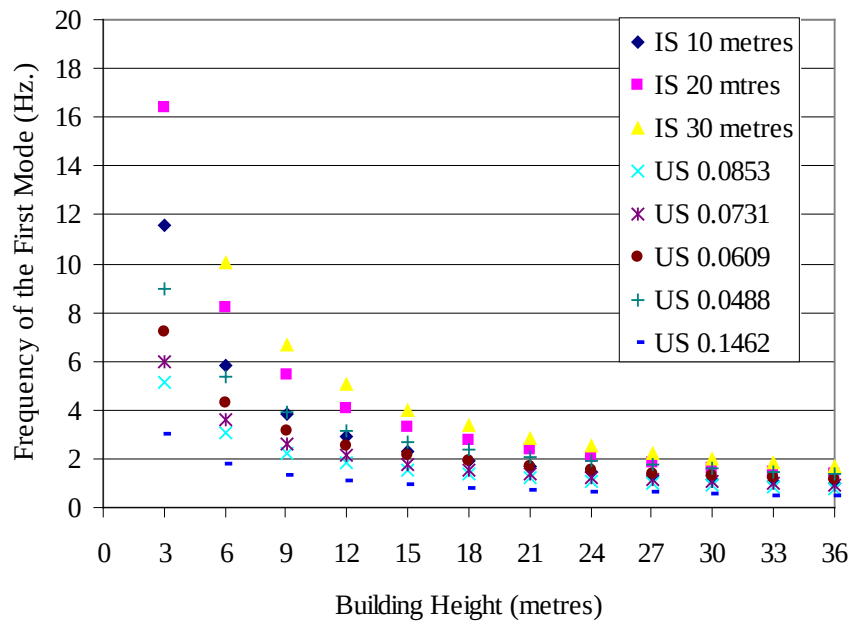


Figure 12. Varying Building Frequency for Different Structural Systems

The mean building period was determined against the height for IS and US data (Figure 13).

¹⁶ The IS represents the IS analysis for three building depths and the US represents the various metric constants used in equation (6).

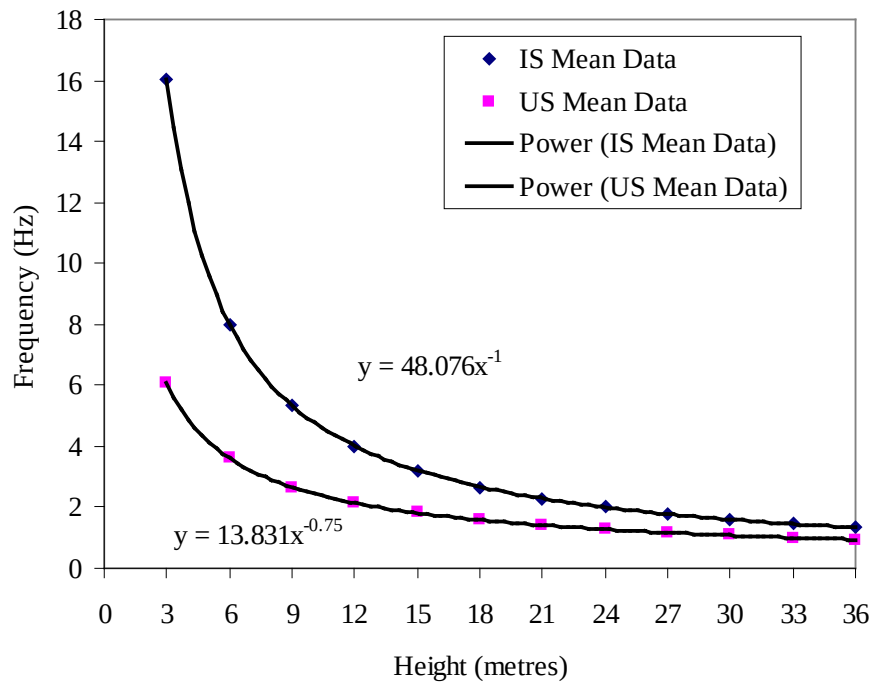


Figure 13. Mean Building Frequency for IS and US systems

The clear result was that the Indian standard provides a higher estimate for the frequency of the first mode than the United States equivalent codes. The difference was not significant for the taller structures, but is significant for short structures. The relationships for the two data sets are shown on the figure. The Indian Standard C coefficient relates the stiffness of the structure to the loading from the earthquake. A stiffer structure attracts proportionally higher load (Fig. 14).

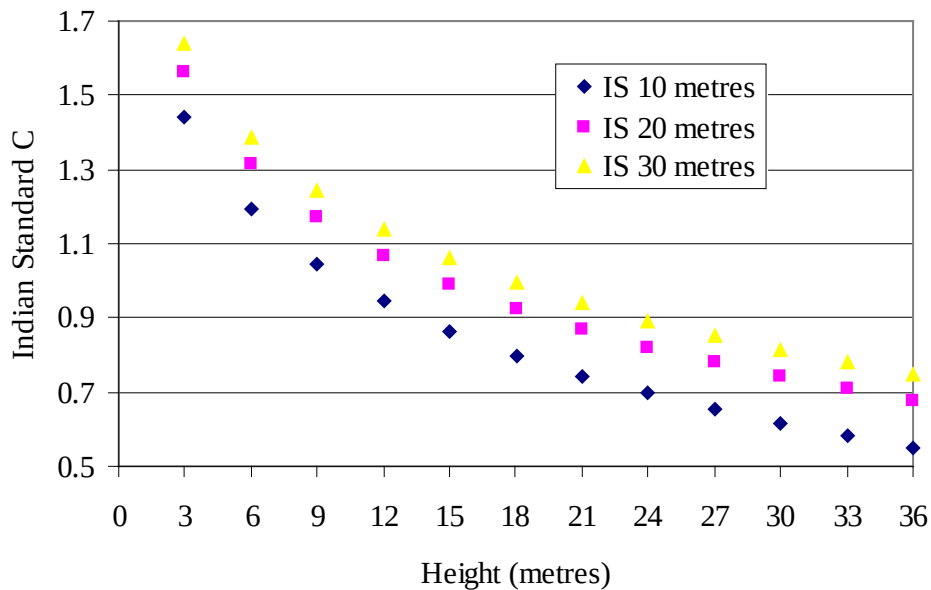


Figure 14. Co-efficient C versus the Building Height for different building depths

The composite factor for the Indian Standard $KC\alpha_h$ was calculated for the buildings (Fig. 15).

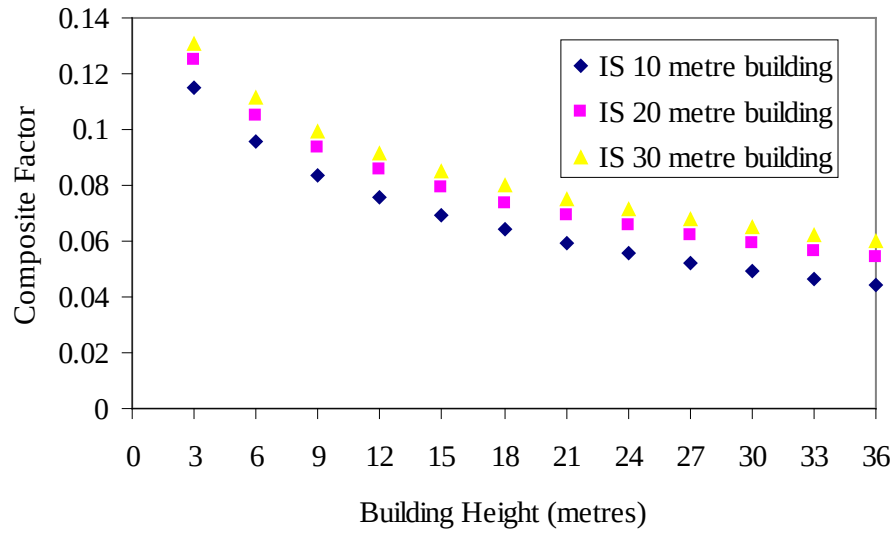


Figure 15. Composite Factor for the three building depths
The composite factor was established for the response spectrum method (Figure 16).

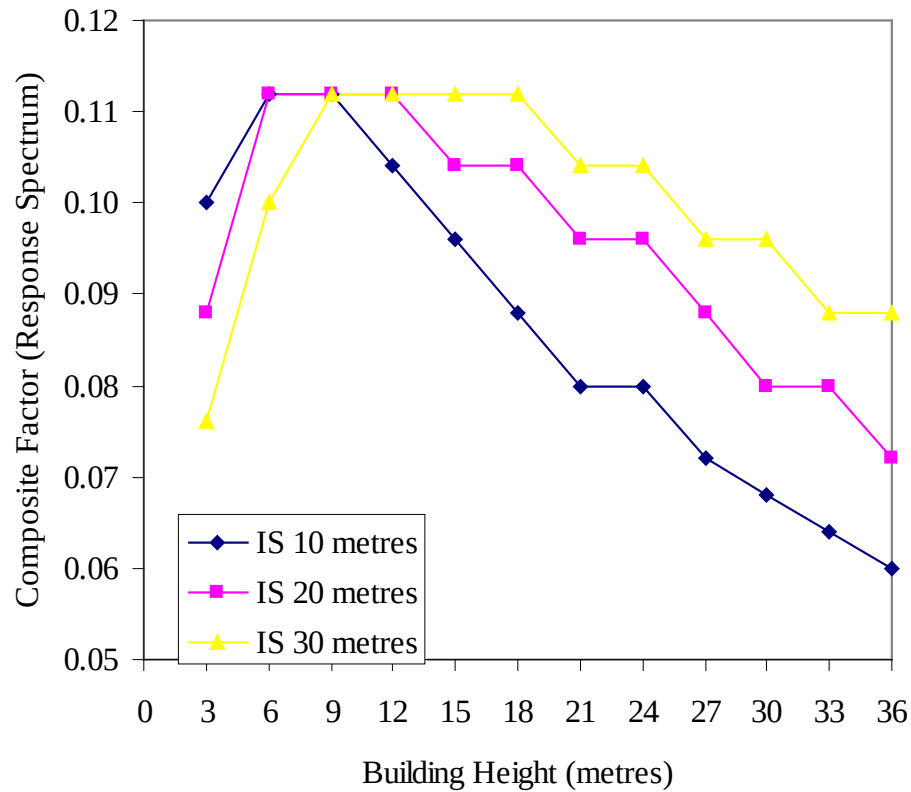


Figure 16. Indian Standard Composite Factor Response Spectrum Method

The composite factor was established for the response spectrum method for the US data (Fig. 17).

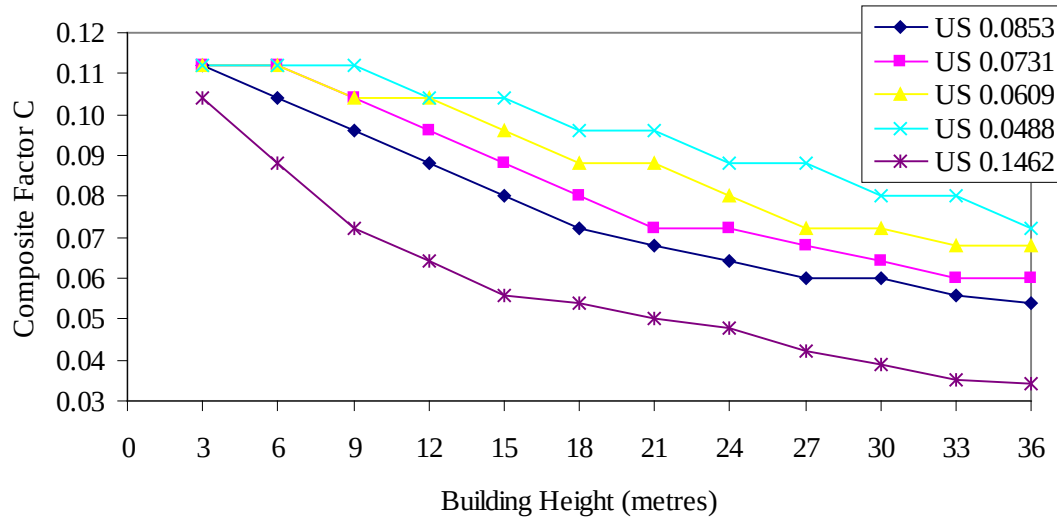


Figure 17. Indian Standard Composite Factor Response Spectrum Method US Data 2 % damping

The base shear equation from ATC 3-06 has the form:

$$V_B = C_s W \quad (9)$$

The composite C_s factor presented in ATC 3-06 Equation 4-1 was calculated assuming a Zone 7 area (equivalent to California at the time) for a response modification factor of 8 and the acceleration factors both 0.4 (Figure 18).

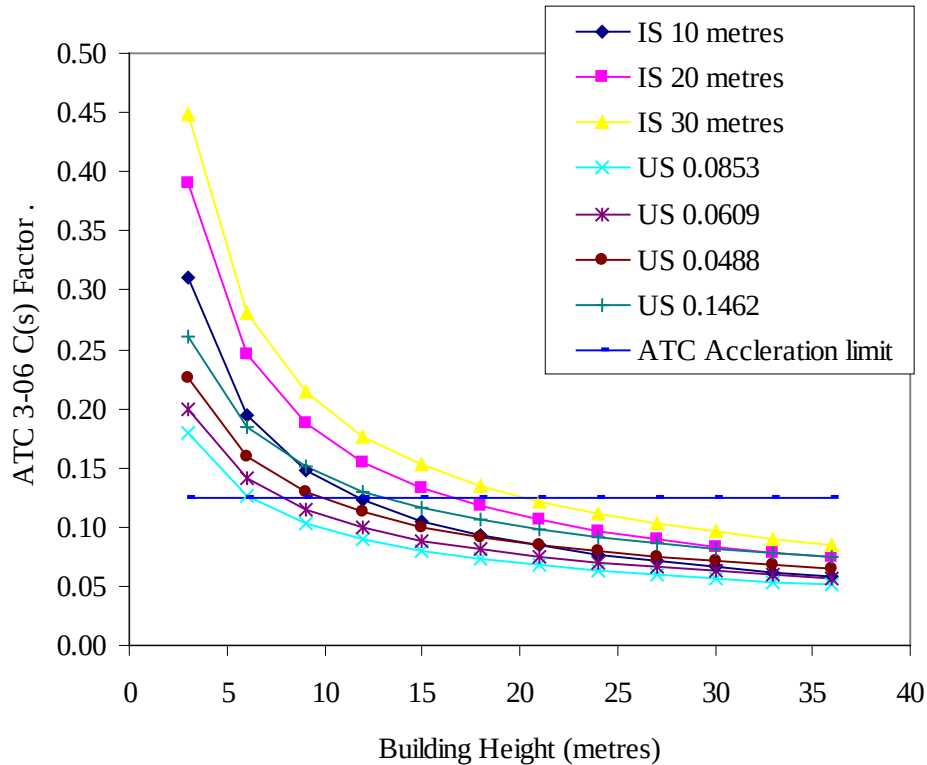


Figure 18. ATC 3-06 composite C_s factor and the ATC acceleration limit

The two controlling features of the ATC method are the acceleration limit that for standard conditions was set by the developers of the method at three Hertz¹⁷. The FFT of the Nahanni earthquake does not suggest that the limit was inappropriate however the two limiting equations in ATC 3-06 are extremely sensitive to the value of R and certainly the value of R selected at eight would appear high simply to bring the composite C_s factor into the range of the results of the other methods. The acceleration limiting composite C_s factor was inversely proportional to R. The base shear equation from FEMA 273 has the form:

$$V = C_1 C_2 C_3 S_a W \quad (10)$$

Where W represents the dead load and the anticipated dead load; C_1 was modification factor to relate expected maximum inelastic displacements to the displacements calculated for an elastic analysis; C_2 represents the modification factor to represent the effect of stiffness degradation and strength deterioration on maximum displacement responses; C_3 represents $P - \Delta$ factor; V represents the base shear load on the structure and S_a represents the response spectrum acceleration at the fundamental frequency of the structure for that direction. The methods for determining the natural frequency of the buildings are based on either modal analysis, equation (4) or for a single storey building a special purpose equation related to displacements of the walls and diaphragm.

The New Madrid Seismic Zone is equivalent in seismic form to the Gujarat State. A review of the NMSZ data can assist in considering the current status of the IS and whether revisions are required to the American Standards. The key points to consider are the results from the Nahanni earthquake, the typical synthetic trace results developed for the NMSZ and the development work by Veletsos and Newmark on the additive impact of pulse in seismic events. The estimated spectral response data for the New Madrid Seismic Zone was determined from the USGS maps in FEMA 273 (Table 7).

Table 7. Spectral accelerations NMSZ

Description	Period 0.2 sec (5 Hz)	Period 1 sec (1 Hz)
MCE 5 % ζ Class B	3	1.23
2 % probability in 50 years 5 % ζ Class B	3	1.23
10 % probability in 50 years 5 % ζ Class B	0.43	0.092

The adjustment of the site class allows for the soil classes A to F. The site class in the NMSZ was assumed D for the purpose of this analysis (Table 8).

Table 8. Spectral accelerations Site Class adjusted NMSZ

Description	F_a	Period 0.2 sec (5 Hz)	F_v	Period 1 sec (1 Hz)
MCE 5 % ζ Class B	1.0	3	1.5	1.84
2 % probability in 50 years 5 % ζ Class D	1.0	3	1.5	1.84
10 % probability in 50 years 5 % ζ Class D	1.45	0.60	2.4	0.221

¹⁷ Calculation using ATC 3-06 equations 4-2 and 4-3 pg 55.

The general response spectrum can be calculated using the FEMA 273 (equations 2-8 and 2-9). Two response spectra are relevant for the New Madrid Seismic Zone; these are the Maximum Credible Earthquake¹⁸ and the 475-year return period¹⁹. The response spectrum (Figure 19) for the NMZS MCE has been developed estimated natural frequencies for the various combinations of building height and type. These spectral accelerations are developed from the USGS data contained in FEMA 273.

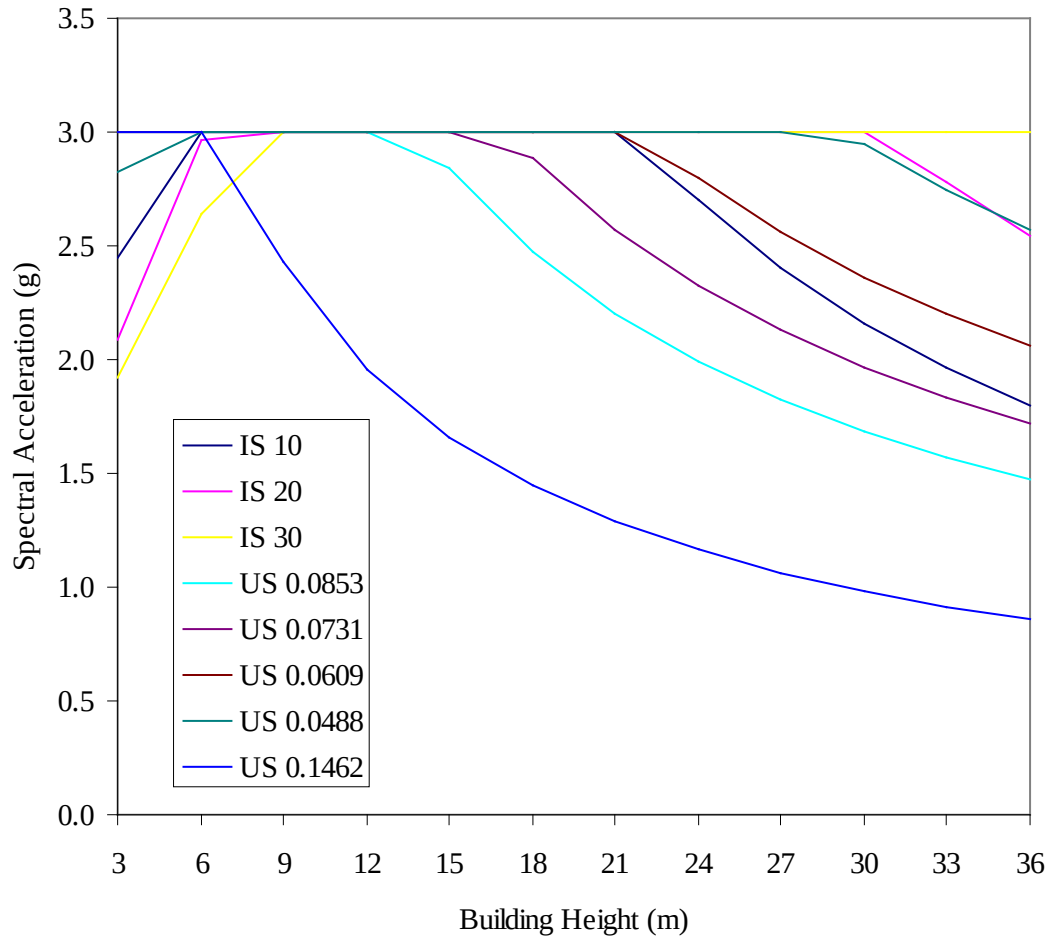


Figure 19. NMSZ MCE Spectral accelerations for the standard building range

¹⁸ The two-percentile event was the MCE.

¹⁹ The return period was used in this chapter as we are discussing characteristic or singular points in the probability space functions that were developed to model the probability of occurrence of the earthquakes. The trivial conversion for this return period was ten percent in fifty years. The difficulty at least for this author from this definition was the building life of five decades. The thought that structures will have a failure of two percent in fifty years or ten percent in fifty years fails to account for the characteristic earthquakes that dominates the fatality counts in earthquakes and that we have general failure in an earthquake of buildings as in the 1988 Spitak earthquake must be considered intolerable in this century. The anomalous situation that occurs in some interplate areas can be explained and the issue of fifty years in these few locations can be considered a trivial issue. It was not considered a trivial issue for areas with characteristic events and one that may fail to account for the basic life safety issues under these specific circumstances.

The problem that can be identified in the use of the standard spectral acceleration shape relates to the occurrence of a tuned structure and FFT for a particular earthquake or simply an earthquake with a spread of energy through the power spectrum. The results demonstrate the scatter in using the different building periods. The collapse of reinforced concrete buildings at a distance of 250 km in Ahmedabad points clearly to the design issue of soft storey as the cause of failure, however without an accurate estimate of the frequency of the building then the base shear can be in error. This bespeaks the interrelated issues of tuned buildings, systematic errors in building frequency and the problem of spikes in the traces of great earthquakes. The purpose of using the NMSZ data was to consider these issues and determine whether the existing analysis methods developed from the earlier work of Veletsos and Newmark (1964) require changes for great or characteristic events.

The response spectrum (Figure 20) for the NMSZ 475 yr return period has been developed for the various combinations of building height and type. These spectral accelerations are developed from the USGS data contained in FEMA 273.

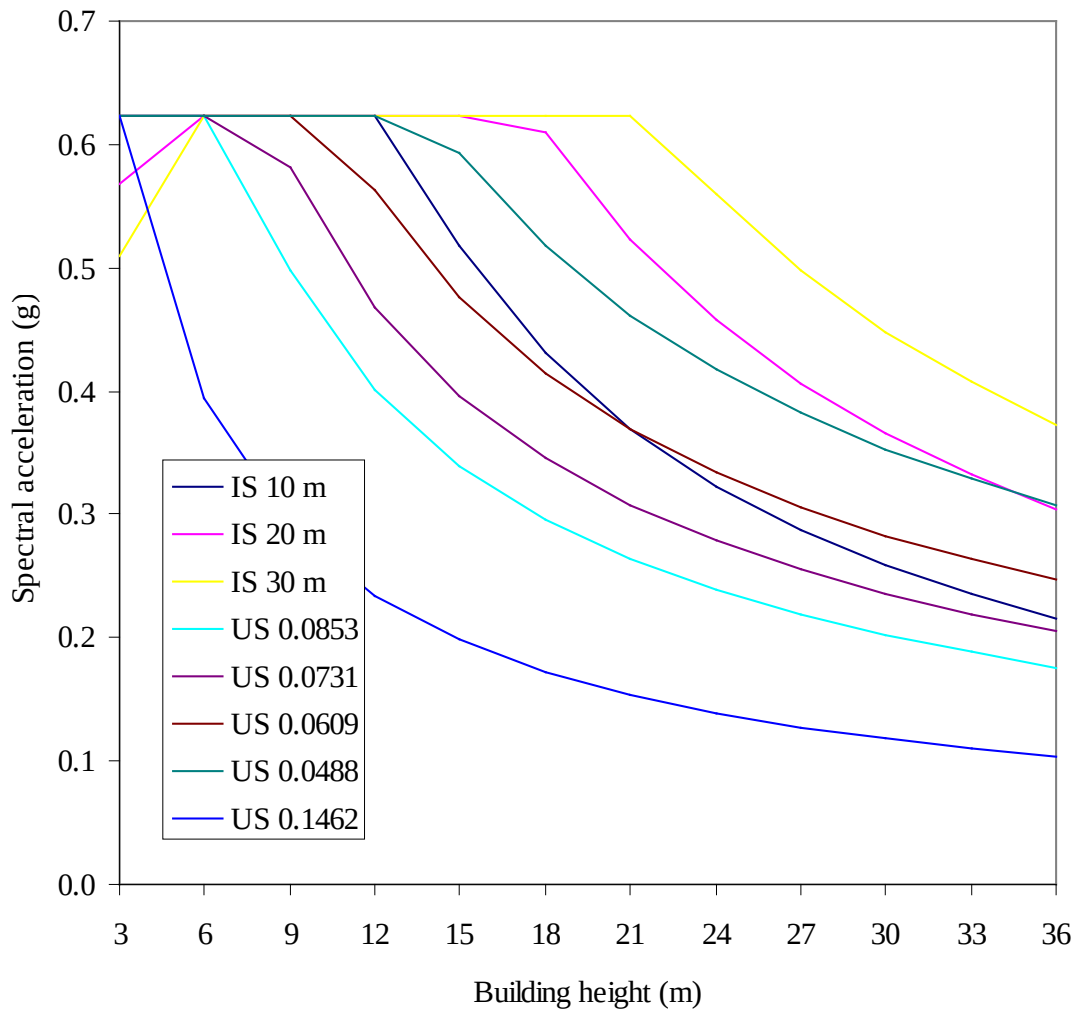


Figure 20. NMSZ 475 year Spectral accelerations for the standard building range

The range in the results between the two codes was interesting. The paleo-seismological data for the NMSZ would suggest that the MCE and the 475-year return period event should be both a great event as defined by Richter. Green and Hall (1994), and Krinitzsky, (2000) discussed in detail the problem of hazard analysis methodologies. The conclusions of these authors are that there are shortcomings in the current analysis techniques.

The method being developed by Kafka and Ebel (2000) provides an alternative method to estimate the seismicity of a regional area. The clear point is 35 percent of all seismic activity in occurs in areas that have not recorded prior activity.

The next step was to consider the differences between the Nahanni earthquake and the various spectra formula.

Veletsos and Newmark (1964) provide the mathematical and numerical derivation of design spectra (shown on their Figure 2.23). The adoption of a constant velocity component to the design spectra represents for their analysis a simple and pragmatic limit to the zero gradient on the spectra for the varying pulse forms. This limit was based on their analysis (Veletsos and Newmark, 1964, § 2.9.2 *Presentation and Discussion of Results, a. Characteristics of the Representative Spectra*) of the form of the design spectra for a range of pulse shapes.

The choice made by Veletsos and Newmark was appropriate for their analysis, of course noting their lack of strong motion data for intraplate earthquakes. The limited strong motion data available now provides data to use the Veletsos and Newmark techniques to allow for acceleration spikes within a normal earthquake trace.

The Nahanni earthquakes FFT and a number of the standard spectral shapes are plotted on Figure 21. The spectral shapes include data from Newmark and Hall (1978, Figure 5) that was developed for the nuclear regulatory agency, the standard NMZS 475 year, a newly developed bounding spectra²⁰ and the Newmark and Hall spectra scaled to 0.6 g's. The frequency has been plotted in the range of zero to ten Hertz as this represents the region of interest for masonry and soft storey RC structures.

The bounding spectrum has been developed to essentially encase the NMSZ earthquakes. The bounding function has been scaled to twenty percent of the normal limits. The function has been developed for comparison purposes only and was not developed for any design purpose and should not be used for that purpose.

²⁰ The bounding spectra has not been based on previous models.

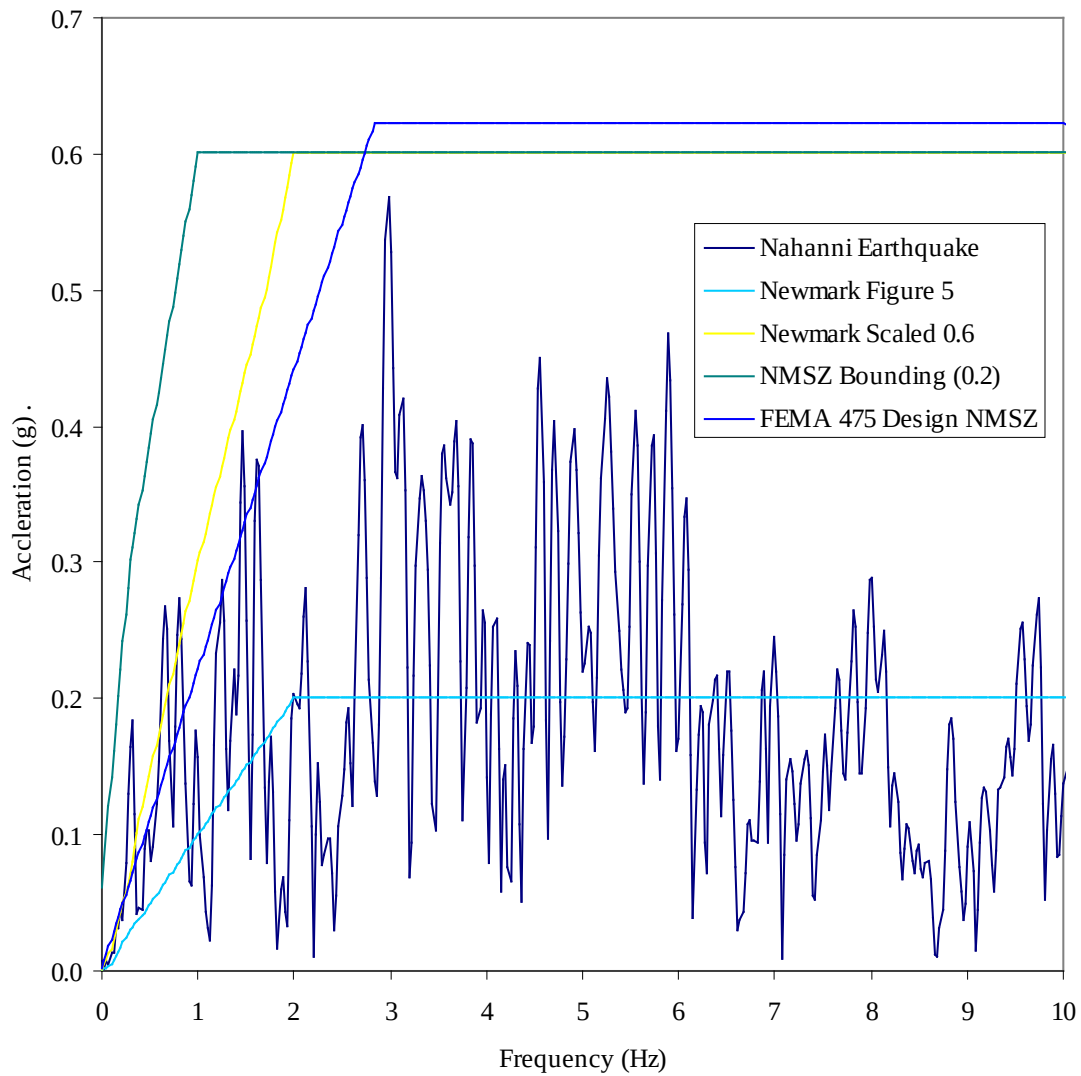


Figure 21. Nahanni earthquake FFT and standard spectra (0 to 10 Hz.)

Only the bounding function does not have peaks in the FFT that exceed the design spectra. The Nahanni earthquake was a M6.8 event of short duration. The data has been recast in the frequency range of zero to two Hertz (Figure 22).

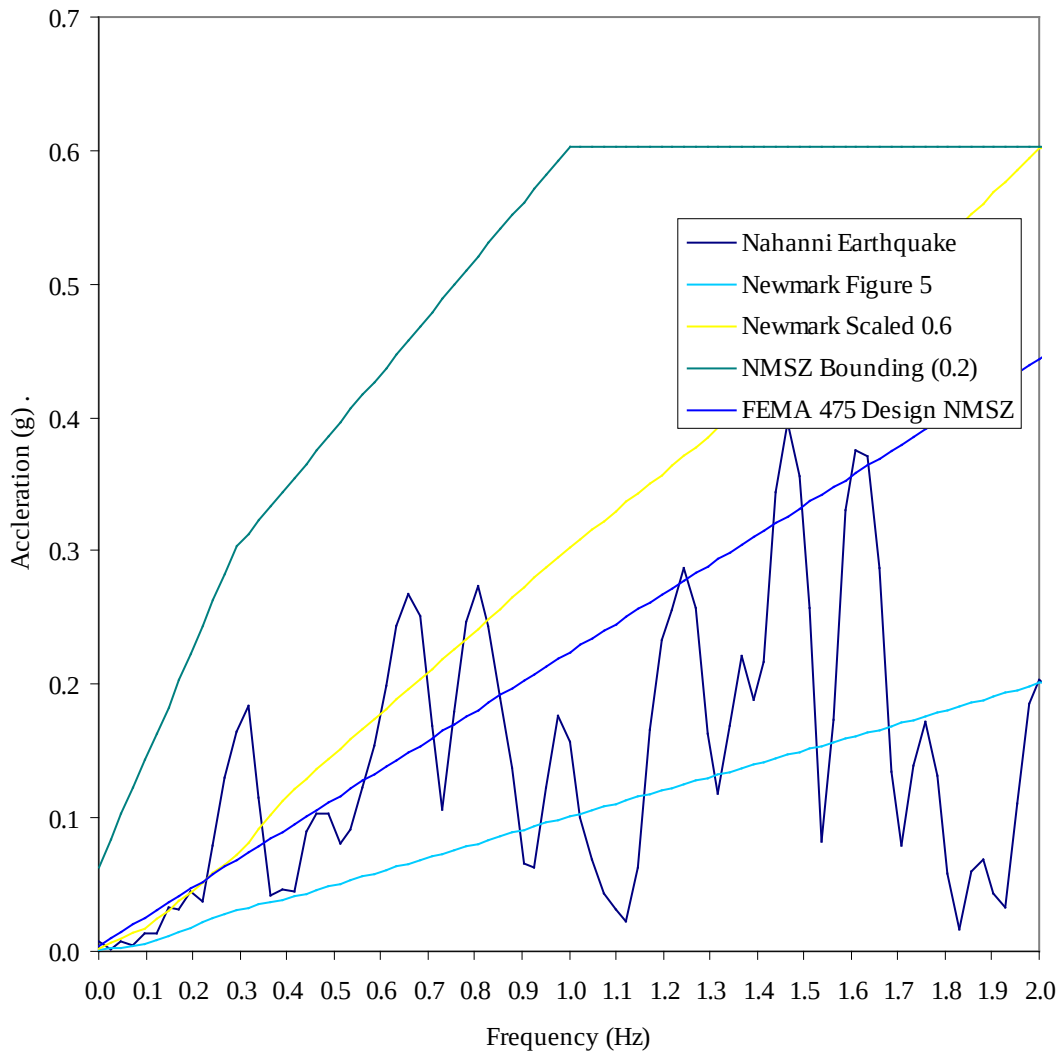


Figure 22. Nahanni earthquake FFT and standard spectrums (0 to 2 Hz.)

The Nahanni earthquake has several peaks in the FFT trace (Figure 22) with numerical values greater than the design spectra. The design spectra are based on the constant velocity region from about 0.2 to 2.0 Hertz (Newmark and Hall, 1978, Figure 5). The Nahanni FFT provides spikes that exceed the design spectra in the range from 0.2 to 2.0 Hertz. What are the impacts of these spikes?

The study into the Loss Assessment in Memphis Buildings had a number of synthetic traces prepared for the Marked Tree region in the NMSZ. The M7.25 earthquake time trace has been converted to the frequency domain using the standard FFT technique. The use of the M7.25 was purely arbitrary. The FFT trace (Figure 23) was approximately damped using the factors derived by Newmark and Hall (1978). The earthquake had a peak intensity of about 3.2g's.

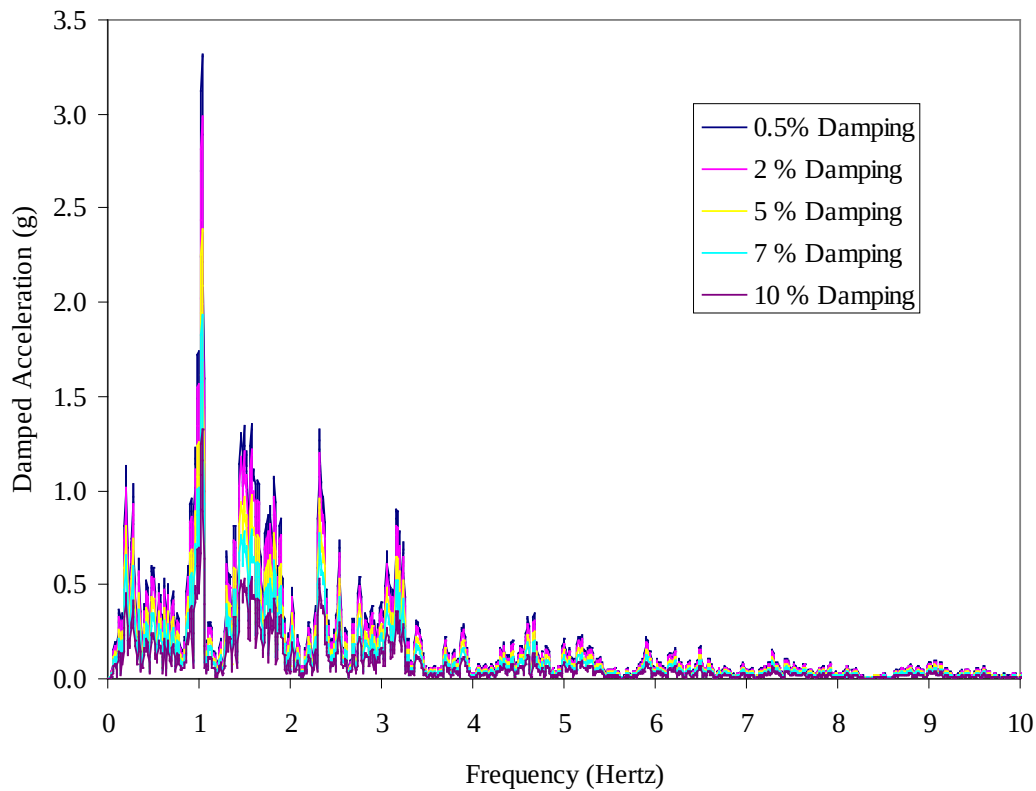


Figure 23. M7.25 Marked Tree earthquake FFT (with damping)

The M7.25 Marked Tree earthquake FFT and a number of the standard spectral shapes are plotted on Figure 24. The spectral shapes include data from Newmark and Hall (1978, Figure 5) that was developed for the nuclear regulatory agency, the standard NMZS MCE, the full bounding spectra²¹ and the Newmark spectra scaled to 0.6 g's. The frequency was plotted in the range of zero to ten Hertz as this represents the region of interest for masonry and RC structures. The FFT for the M7.25 breaks the design spectra at a number of low frequencies. The NMSZ MCE will require significant damping in the building to ensure that demand does not exceed capacity.

The data has been recast in the frequency range of zero to two Hertz (Figure 25) and with the dependent variable the period (Figure 26). The spectral shapes have been tabulated for the standard spectra and the estimated bounding spectra (Table 9). Figure 27 and Figure 28 shows the period versus frequency graph, and the various design spectra used in this report for the ranges of 0-100 Hz and 0-10 Hz. The purpose of showing the period graph was to demonstrate the rapid slope changes in this function in our region of greatest interest.

Queste (2000) has shown a method to allow for the variation in the peaks in the SDOF operator for the earthquake record CH85VALP.070. The method uses a shift in the curve and the intersection frequency (Figure 29). This method for allowing for the spike represents the same mathematical procedure as the development of the bounding function in the frequency plots.

²¹ The bounding spectra has not been based on previous models.

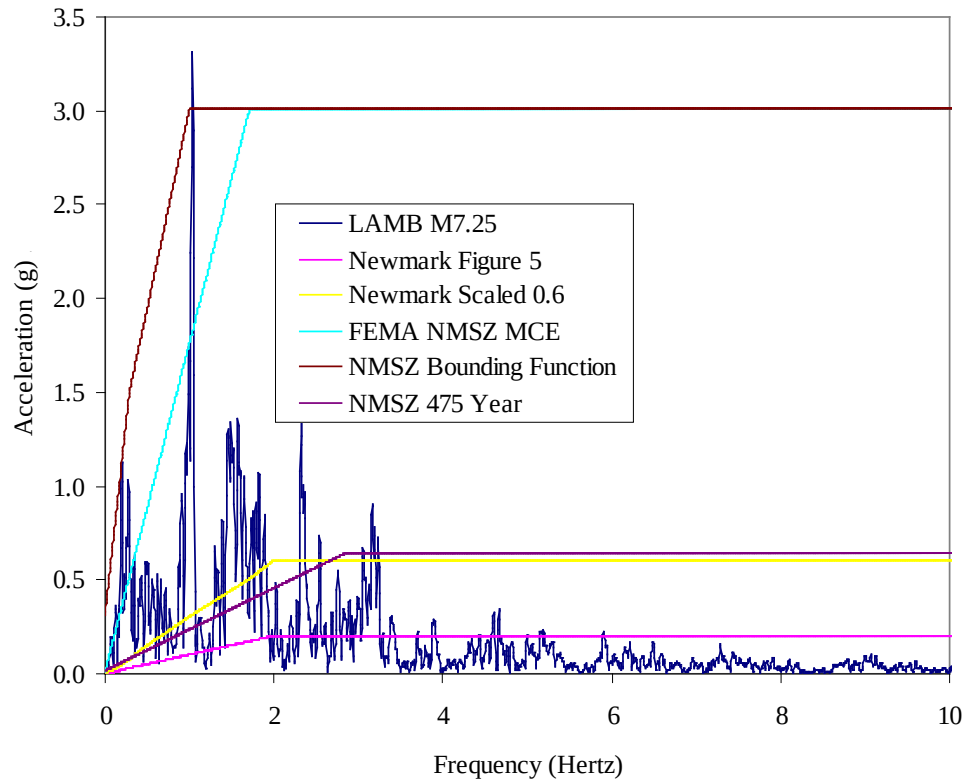


Figure 24. M7.25 Marked Tree earthquake FFT with standard spectral shapes

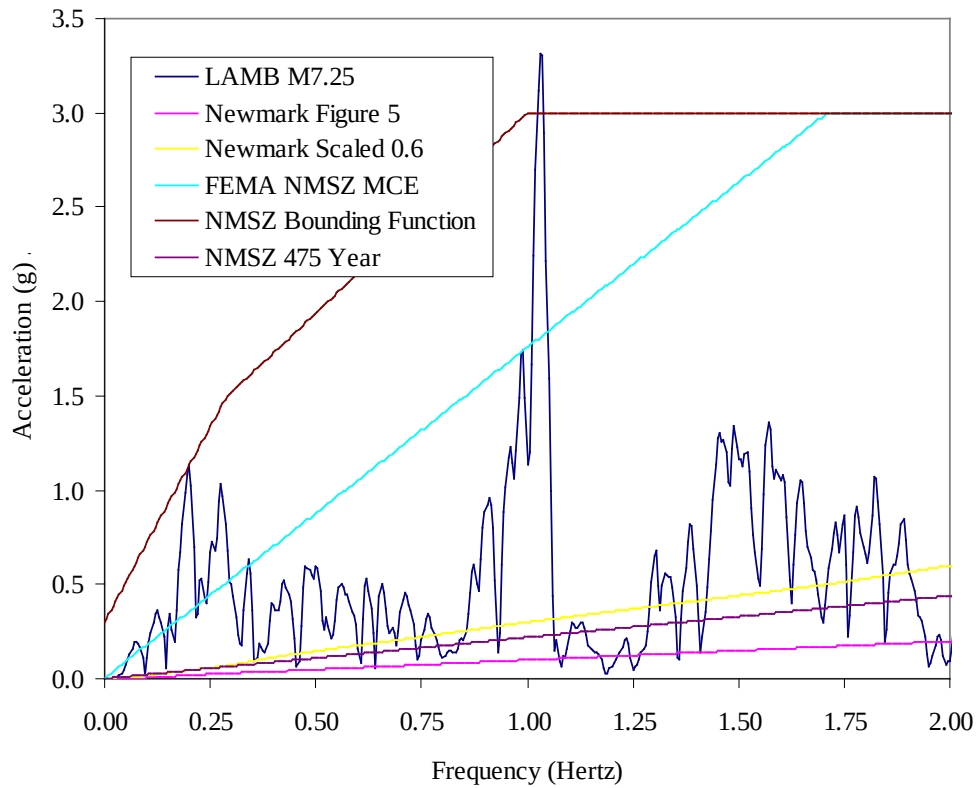


Figure 25. M7.25 Marked Tree earthquake FFT with standard spectral shapes 0 to 2 Hz.

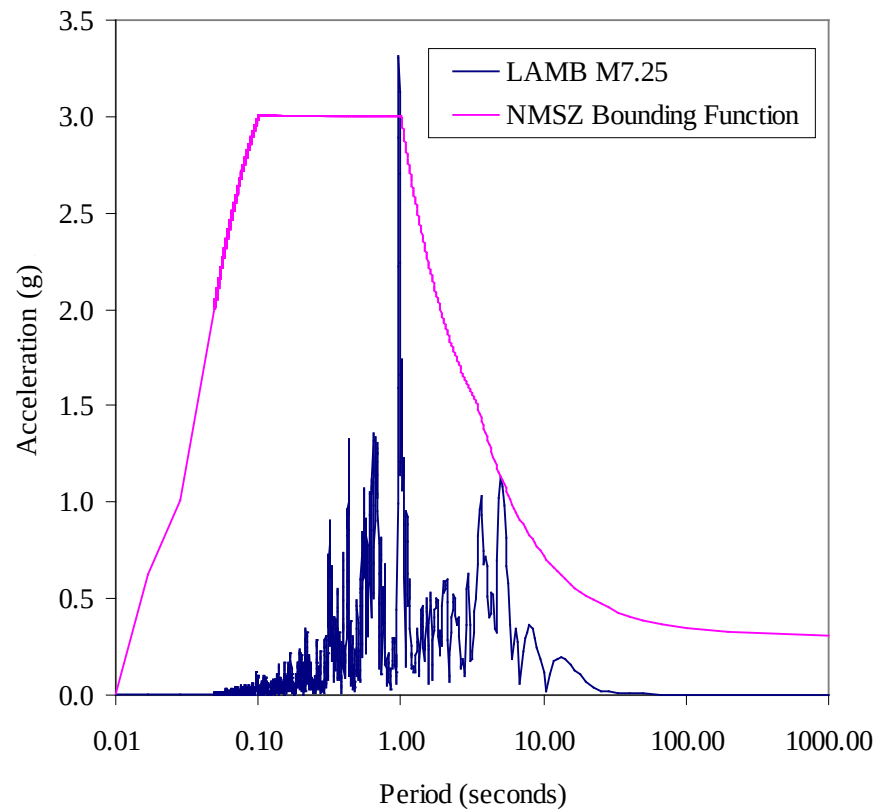


Figure 26. M7.25 Marked Tree earthquake FFT with the bounding shape.

Table 9. NMSZ Tabulated Spectral Points

Frequency Hertz	Period seconds	Acceleration (g)					
		Newmark Figure 5	Newmark scaled 0.6	NMSZ MCE	LAMB NMSZ Bounding	LNB (0.2)	NMSZ 475 year
0.000	1000.000	0.000	0.000	0.000	0.30	0.06	0.00
0.098	10.204	0.004	0.015	0.172	0.70	0.14	0.02
0.293	3.413	0.030	0.070	0.514	1.50	0.30	0.06
0.415	2.410	0.040	0.120	0.728	1.76	0.35	0.09
0.610	1.639	0.060	0.180	1.071	2.17	0.43	0.13
1.001	0.999	0.100	0.300	1.757	3.00	0.60	0.22
1.709	0.585	0.170	0.500	3.000	3.00	0.60	0.38
2.002	0.500	0.200	0.600	3.000	3.00	0.60	0.44
2.82	0.355						0.62
10.010	0.100	0.200	0.600	3.000	3.00	0.60	0.62
35.010	0.029	0.170	0.500	1.580	1.00	0.20	0.38
60.010	0.017	0.170	0.170	1.340	0.62	0.12	0.21
100.000	0.010	0.000	0.000	1.200	0.00	0.00	0.00

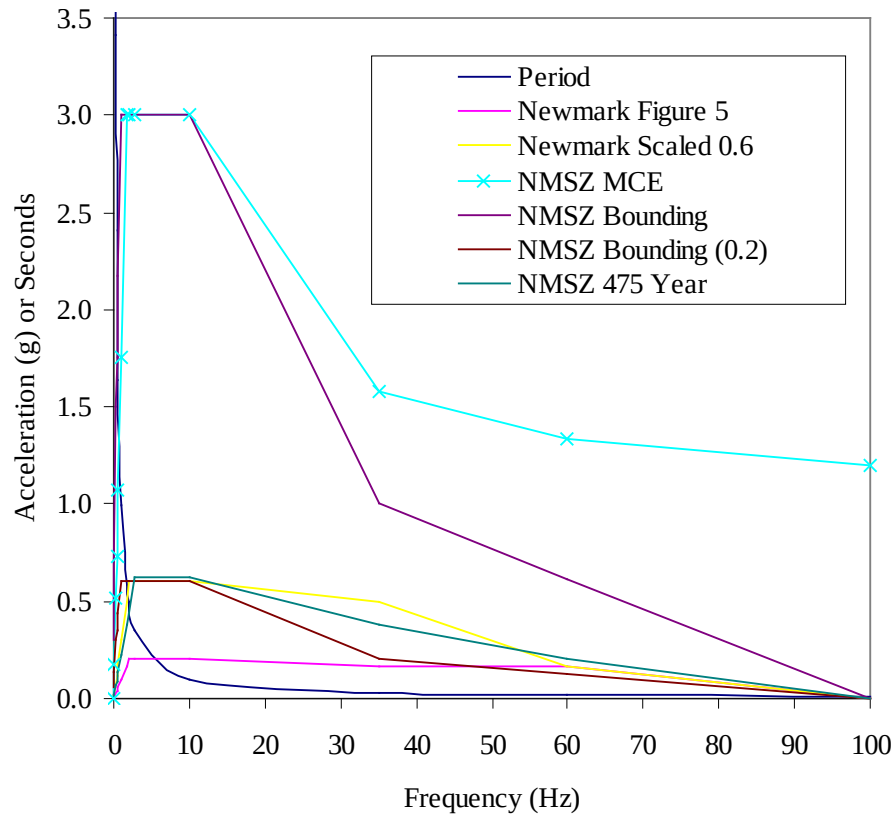


Figure 27. Period versus Frequency and the various design spectra

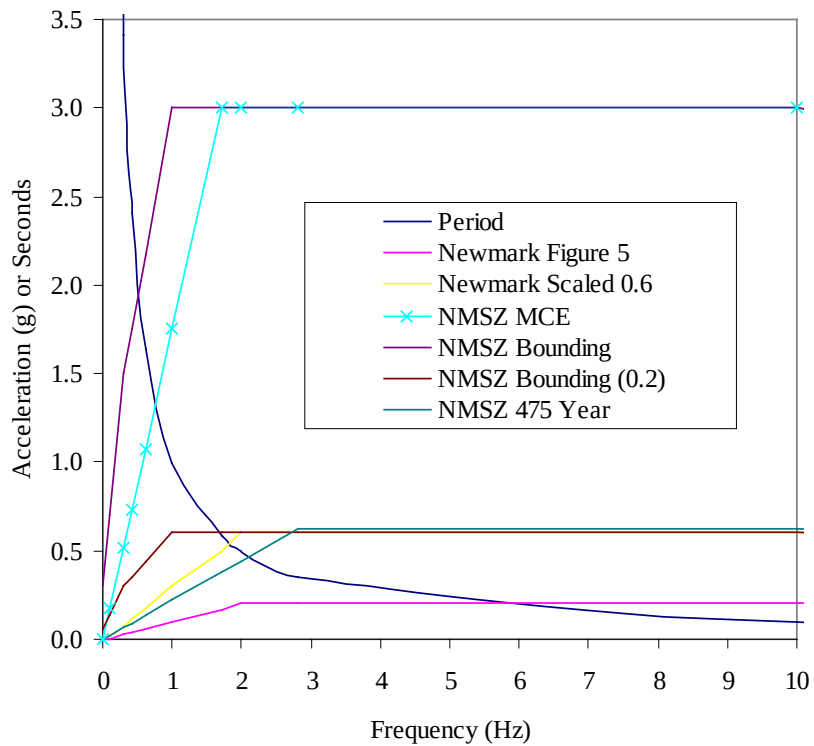


Figure 28. Period versus Frequency and the various design spectra 0-10 Hz.

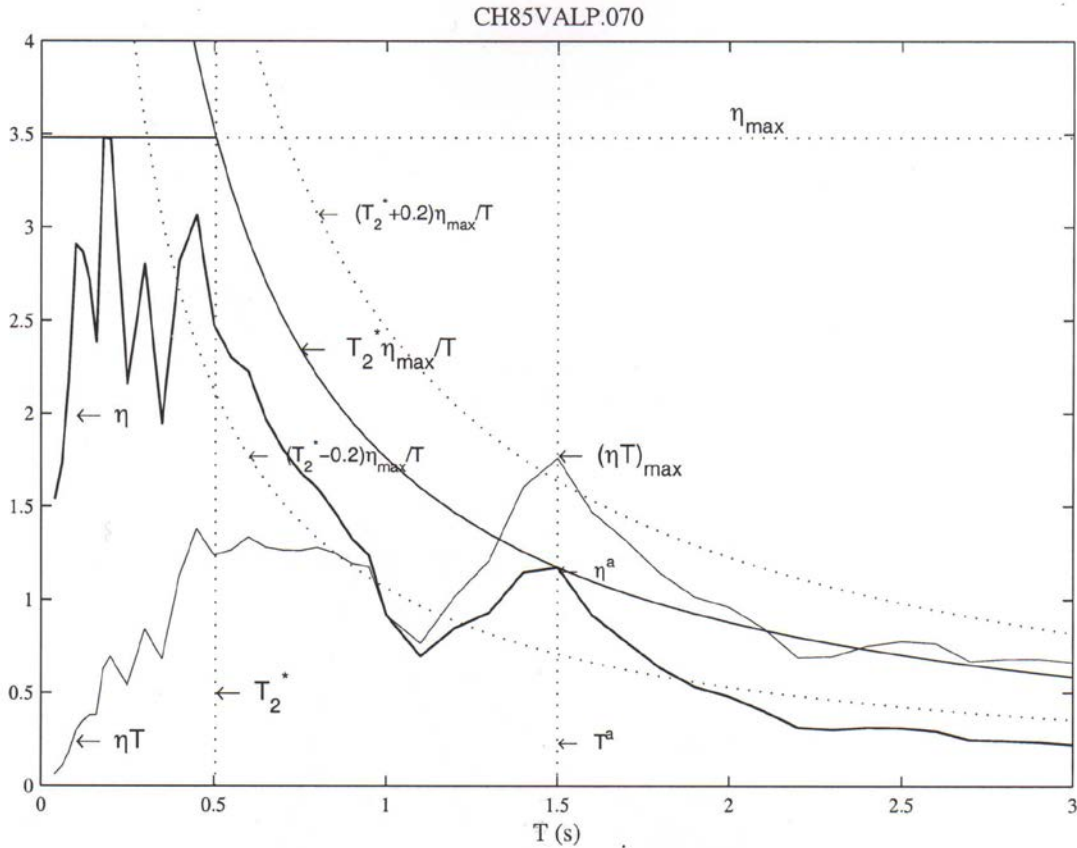


Figure 5.6: Graphical determination of the period T_2^* for the CH85VALP.070 record

Figure 29. CH85VALP.070 Method to adjust for a spike in the record.

A number of earthquake details are provided in Table 10. These events are typical design earthquakes used in the development of standards.

Table 10. Earthquakes, Location, Date, Magnitude, and Duration

Earthquake Location	Recording Location	Date of the event	Magnitude	Duration
Marked Tree Arkansas	Synthetic		7.25	40
Nahanni (1)	Station 1, Iverson, Canada	23 Dec, 1985	6.8	12
Nahanni (2) St 3,	Battlement Creek, Canada	23 Dec, 1985	6.8	20
Saguenary (1) St 8,	La Malbaie, Ca,	25 Nov, 1988	5.4	20
Saguenary (2) St 16,	Chicoutimi Nord PQ, Canada	25 Nov, 1988	5.4	15
Irpinia Calatrini,	Italy	23 Nov, 1980		70
Newcastle Australia	Synthetic	28 Dec, 1989	5.5	6
Miramichi Loggie Lodge,	Canada	6 May, 1982		1
Imperial Valley (1)	El Centro, USA	18 May, 1940	7.1	30
Imperial Valley (2)	El Centro Pknold,	USA 18 May, 1940	7.1	28
Miyagi-Ken-Oki,	Sensai City, Japan	12 June, 1978,		35
Parkfield California		27 June 1966		20
San Francisco,	Cholame St, California	22 March, 1957		10
Kern County	Taft High, California	21 July, 1952	7.6	40

The first peak and the maximum peak in the FFT spectra for the earthquakes listed in Table 10 are shown in Figure 30.

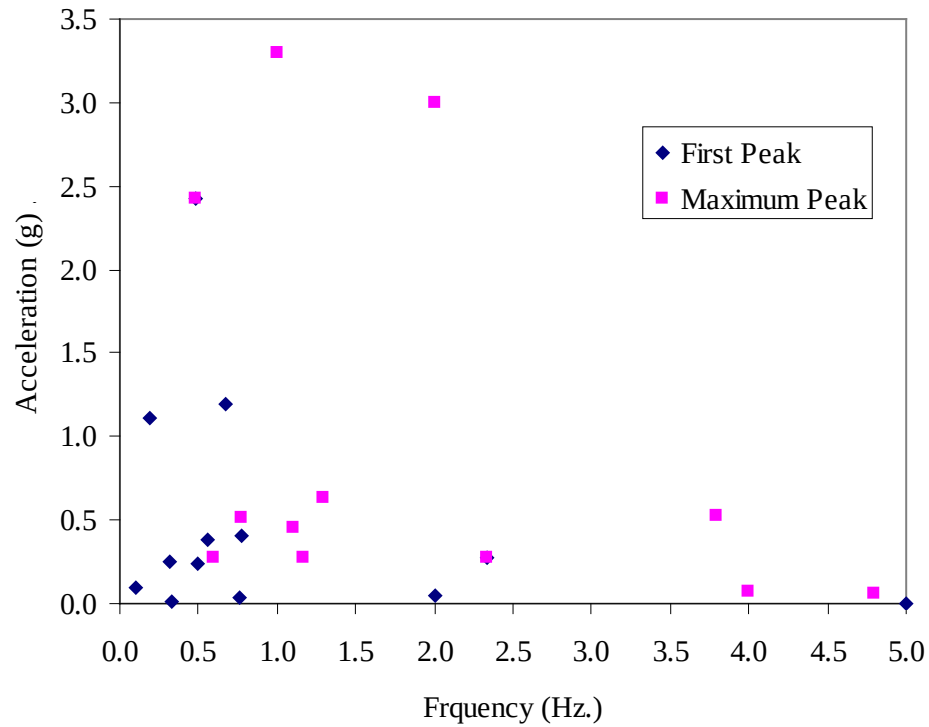


Figure 30. First and Maximum Peaks in the Frequency Spectra.

Veletsos and Newmark (1964) investigated the spectra for a variety of pulse loadings for bomb blasts. The research work of Veletsos and Newmark included a comparison of the results to the El Centro and the Eureka²² earthquake SDOF spectral shape. These two researchers commented on the area of constant peak velocity as being the controlling feature from about 0.3 to 3 Hertz. This comment was true for the interplate earthquakes that were studied at the time for their report. Additional data having been collated since that time (1964) provides some further guidance as to the controlling features in the range of 0.3 to 3 Hertz. The frequency range of real interest for our work was the 1 to 2 Hertz range.

Veletsos and Newmark (1964) provide extensive guidance on handling the types of responses that were recorded for the Nahanni earthquake and that have been synthetically generated for the Marked Tree event. The first and maximum peaks in the FFT spectra have been plotted in Figure 30 for the earthquakes listed in Table 10. The frequency range of 0 to 1 Hertz has been plotted in Figure 31 to show the large earthquakes that have a high acceleration in the frequency range of 0 to 1 Hertz.

²² Eureka earthquake of December 21, 1954 S11E Component.

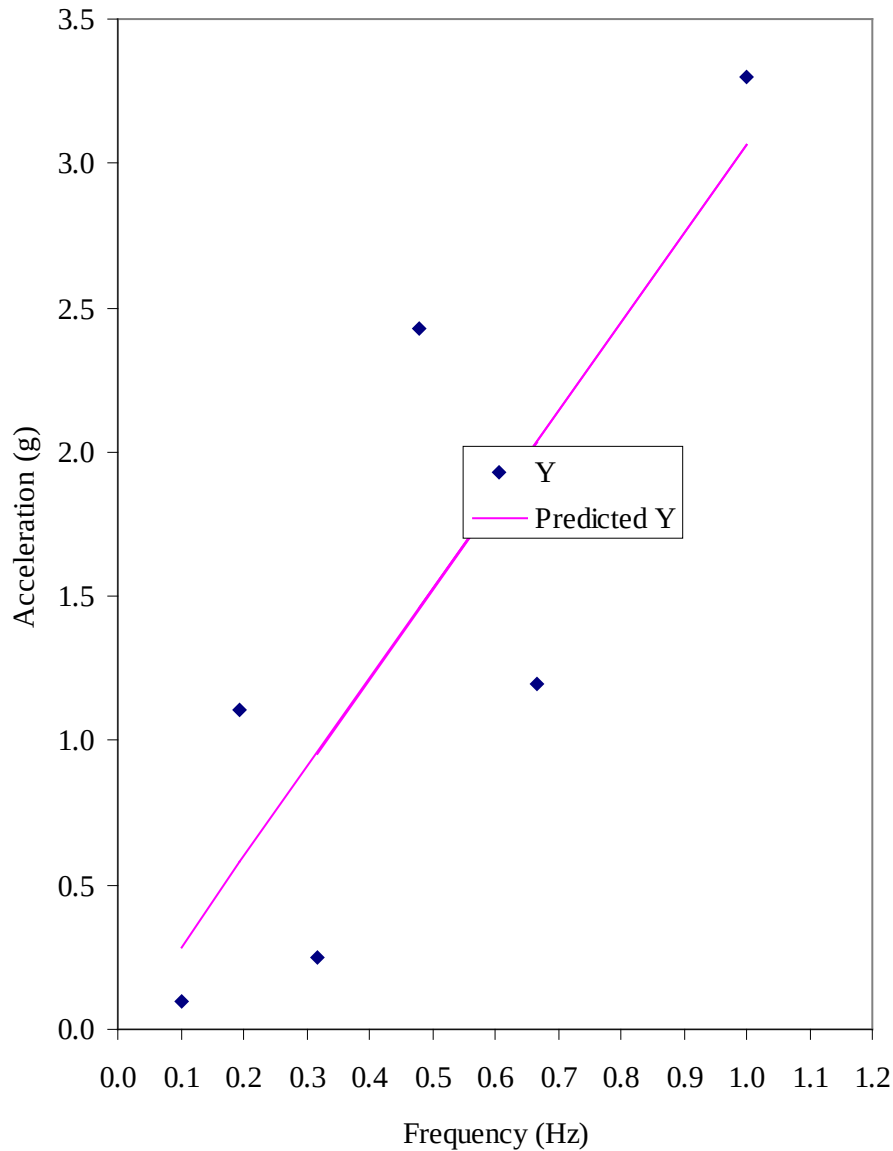


Figure 31. First and Maximum Peaks in the Frequency Spectra 0 to 1 Hertz.

Whilst there was limited data to make any firm observations the analysis completed by Veletsos and Newmark (1964) and the further data on these larger intraplate events provides some guidance on the applicability of the standard spectra shapes. The applicable comments are:

1. The interplate earthquakes such as El Centro contain a range of frequencies that provide for a series of peaks in the spectrum for the 0.3 to 3 Hertz region creating the constant velocity section of the standard spectrum. Veletsos and Newmark (1964) clearly show that this region was dominated by the acceleration and the velocity components rather than the velocity component alone.

2. The Nahanni and similar earthquake events include large pulses in the time trace that dominate the spectral shape. These pulses in the Nahanni earthquake are a 1 Hertz underlying pulse with a non-damped acceleration of about 0.6 g's it will not be significantly reduced by low damping levels (0.02 to 0.05), and a 6-Hertz pulse at about 10 seconds with a peak acceleration of 1.5 g's. This pulse at 6 Hertz will be damped out in the sample damping range (0.02 to 0.05).
3. There are no physical reasons that the spectral shape from 1 to 3 Hertz has a constant velocity component and the FFT data provided in Figures 30 and 31 suggest that the asymptote can be reached in these larger intraplate events at 1 Hertz. The region of constant velocity was simply an overlapping area of asymptotes for the El Centro and other earthquakes. The large intraplate earthquakes may contain pulses. The analysis from Veletsos and Newmark (1964) for pulses provides a simple method to deal with pulses in an otherwise fairly standard earthquake signal.
4. The Figure 5 from the Newmark and Hall report (1978) and all derivative curves developed since then should be re-examined in detail to consider the potential impact in intraplate areas of these large low frequency and higher frequency pulses.
5. The standard curve with a constant velocity component will require modification to return to the asymptote form.²³

In conclusion the Indian Standard 1893: 1984 whilst generally maintaining the principles outlined in ATC 3-06 has selected different base equations for the natural frequency of the structures, the IS method would appear to provide more conservative answers for the frequency of the structure.

Estimation of the composite C_s factor that directly and linearly relates the mass of the structure to the base shear has been established using two methods in the IS. The two methods are the seismic co-efficient method and the response spectrum method. These two methods provide peak acceleration for Zone V of the Indian continent of 0.13g's for a two percent damped structure and about 0.38 g's for zero damped structure.

The Indian Standard 1893: 1984 will require amendment after the Gujarat earthquake as the equivalent Australian standard was amended after the Newcastle earthquake and the US Standards after the Northridge earthquake and to permit incorporation of the seminal work completed by FEMA, NSF and others in the past decade. The main amendments will need to include a revision to the method of calculating the composite C_s factor that for the Zone V in the Indian Standard would appear to provide for rapid attenuation and to require amendments to the spectrum response data that forms the basis of the code. The US data for the New Madrid Seismic Zone appears the applicable data to use to supplement the data obtained from the Gujarat earthquake.

That the Engineering Community in the United States uses period and US customary units in the day-to-day analysis of structures would appear in the long run to be an untenable position. In completing this analysis between the two standards, the translations and chances for error were significant simply because the base data for seismicity is collected in the SI units and is measured in units of seconds. This data has a linear translation to frequency, whereas the period provides an inverse measure that squashes the collected data in an untenable form. The chance for error grows

²³ Thanks to Professor W Hall for suggesting where to look in this data and for the helpful comments on the constant velocity region of the standard spectra.

in this continued use of these methods and fundamentally breaches the ASCE Code of Ethics Canon One for preservation of life. The suggestion that the method is that preferred by the Contractors raises an interesting argument heard frequently in this matter, which self-evidently fails the Code of Ethics test.

The work of Veletsos and Newmark (1964) should be examined with the additional data to determine if there are suitable changes to the standard spectral shape for these larger intraplate events. Finally the natural frequency formula should be reviewed to determine the appropriate set of standard equations. The use of a method of artificially changing the frequency to provide some conservatism to the base shear estimates should be stopped and if further conservatism was warranted it should be addressed at the appropriate equations and not through the period formula.

The use of US Customary units and period should be discouraged in earthquake engineering. The simple reasons are to be in line with seismological practice and general world practice.

5. Building Repair Standards

IS 13935: 1993 Repair and Seismic Strengthening of Buildings – Guidelines The Bureau of Indian Standards is responsible for the

6. Concrete Standards

IS 456: 2000 Plain and Reinforced Concrete Code of Practice (Fourth Revision), and IS 13920: 1993 Ductile Detailing of Reinforced Concrete Structures Subjected to Seismic Forces - Code of Practice

7. Steel Construction

IS 800: 1984 Code of Practice for General Construction in Steel (Second Revision) was developed in parallel with the equivalent AS 1280 Steel Structures Code from Australia. This code draws on an extensive body of work by Trahair and Hawkins from the University of Sydney in the late 1960's. The code simply requires revision to bring it in line with modern codes of practice.

The code by way of example limits compressive stresses to a level determined from geometrical properties.

8. Masonry Standards

IS??????? Masonry Structures The specific objectives are to review the Indian standards related to earthquake engineering in relation to the United States codes of practice. The review provides an opportunity to consider both the Indian codes and the US codes in light of this major intraplate event.

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